Breaking Random-Init Symmetry: Theory-Informed Initialization for ReLU Networks

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Abstract

Standard random initialization is permutation-symmetric: every assignment of neuron indices to functional roles is equally likely, so a trained network is one of many symmetry-equivalent representatives. We propose THEORY-INIT, an initialization that selects a single canonical representative by construction. Placing deterministic rows in fixed first-layer slots, propagating them through middle layers of the form $I + \Delta$ with small Gaussian Δ, and reading out with a small-random output layer breaks the permutation symmetry directly at initialization, independent of which deterministic rows are used. When those rows additionally encode *domain theory* of the target, the same construction produces an accuracy advantage on physics-rich tasks. We give a theoretical account of why indices are approximately preserved through ReLU layers, with a moment identity, a worst-case L^2 bound, and a high-probability near-isometry of the middle-block Jacobian. Empirically, across four regression tasks and a small CNN on MNIST and CIFAR-10, two THEORY-INIT networks *at initialization* sit an order of magnitude closer in Frobenius distance than He pairs and remain measurably closer after training, and Git Re-Basin returns the identity permutation between two THEORY-INIT networks but a non-trivial one between two He networks; the pre-alignment signature replicates on the small CNNs, though strict identity matching does not. THEORY-INIT thus offers a route to weight-space symmetry removal that is built in rather than searched for.

1. Introduction

A trained ReLU network and any neuron-permuted version of it compute the same function (Hecht-Nielsen, 1990; Entezari et al., 2022), so every trained network is one representative of an enormous permutation orbit. Standard initialization (He (He et al., 2015), Xavier (Glorot & Bengio, 2010), orthogonal (Saxe et al., 2014)) is permutation-equivariant: each neuron-permuted version of a draw has the same density, so the init commits to no particular orbit representative and leaves the choice to the optimizer. Aligning two such trained networks therefore requires explicit search, and a substantial line of work builds on mode-connectivity results (Garipov et al., 2018; Draxler et al., 2018; Frankle et al., 2020; Entezari et al., 2022) to perform or sidestep that search: weight-matching and model merging (Singh & Jaggi, 2020; Ainsworth et al., 2023; Jordan et al., 2023; Theus et al., 2025; Stoica et al., 2024), weight-space averaging of fine-tuning runs from a shared pre-trained start (Wortsman et al., 2022; Sadrtdinov et al., 2023), and lottery-ticket mask transfer across initializations (Frankle & Carbin, 2019; Adnan et al., 2025).

Symmetry-removing architectures. A complementary line reduces parameter symmetry by changing the architecture itself: Lim et al. (2024) build *Asymmetric Networks* with provably fewer parameter-space symmetries, either by freezing weights to large fixed random constants (W-Asymmetric) or by a fixed-mixing gated nonlinearity (FIGLU, σ-Asymmetric), and show alignment-free linear mode connectivity at CIFAR and citation-graph scale. Their study removes the symmetry group for the whole weight space; THEORY-INIT instead keeps a standard ReLU network and its full permutation group intact and selects one canonical orbit representative at initialization. The two are duals: they act on the architecture, we fix the gauge at $t{=}0$ with the architecture, optimizer, and trainable-parameter count unchanged, which is why our prescribed rows carry interpretable content at small magnitude rather than serving only as large fixed constants. See Section E.1 for the detailed comparison.

We ask the converse question: if we already know something about the target function (a Crippen atomic-contribution model for molecular LogP, an Abrams cement-water law,

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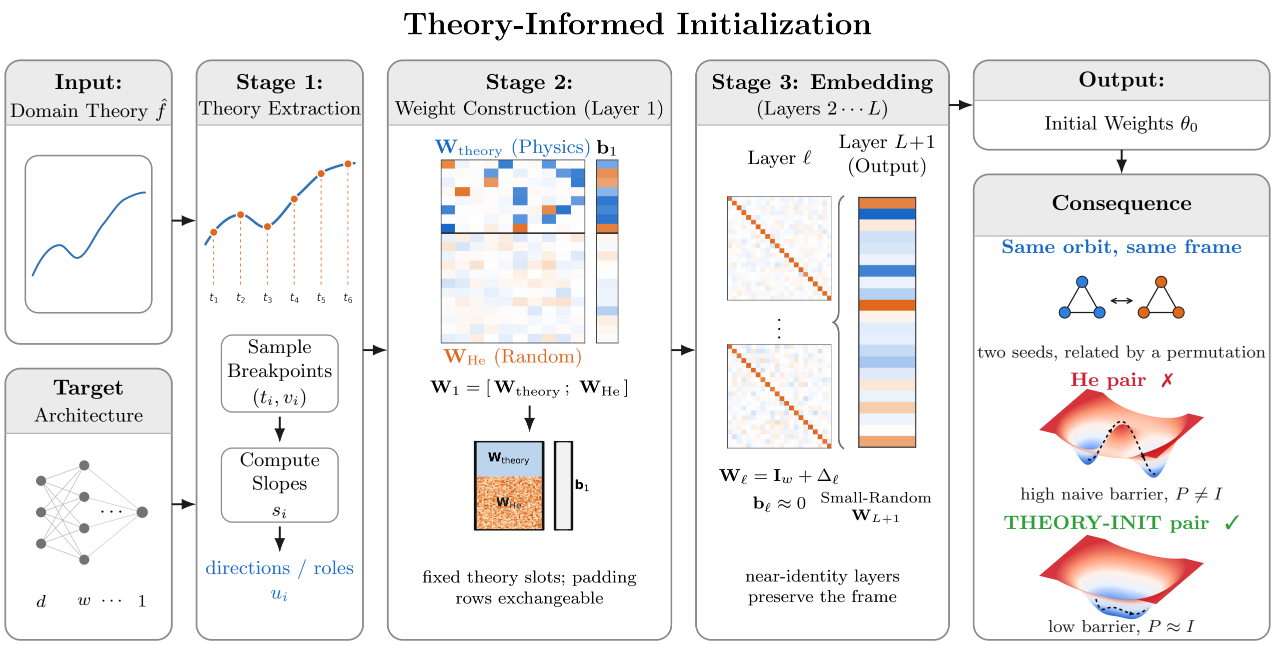

Figure 1. **THEORY-INIT overview.** Stage 1 extracts breakpoints and slopes from a domain theory $\hat{f}$; Stage 2 writes them into fixed first-layer slots $W_1 = [W_{\text{theory}}; W_{\text{He}}]$ with small-magnitude He padding (the theory fraction is enlarged for visibility); Stage 3 propagates them through near-identity middle layers $W_\ell = I_w + \Delta_\ell$ and a small-random readout. Two THEORY-INIT seeds share an orbit representative; two He seeds do not.

a Froude-number scaling for hull resistance), can we use that knowledge to select a canonical point in weight space at $t{=}0$ rather than aligning trained networks post hoc? Existing methods that inject prior knowledge work elsewhere: identity-style initializations (Pan et al., 2025; Zhao et al., 2022) start at a symmetric point but carry no domain theory; physics-informed networks (Raissi et al., 2019) and Sobolev training (Czarnecki et al., 2017) act through an auxiliary loss; curriculum, warmup, and pre-training on theory-generated surrogates act through optimization. None bind specific neuron indices to specific functional roles.

Linear mode connectivity. Two trained networks θ_A, θ_B are *linearly mode-connected* (Frankle et al., 2020) when the linear interpolation $\theta(\alpha){=}(1{-}\alpha)\theta_A{+}\alpha\theta_B$ stays at near-baseline loss; the worst excess loss along the path is the *barrier*. Entezari et al. (2022); Ainsworth et al. (2023) empirically showed that the (often large) naive barrier between independently trained networks is substantially reduced or removed once θ_B is brought into the same neuron-permutation frame as θ_A, defining the post-alignment barrier $\mathcal{B}_{\text{align}} = \min_P \mathcal{B}_{\text{naive}}(\theta_A, P{\cdot}\theta_B)$. A small aligned barrier means the networks lie in symmetry-equivalent corners; a small *naive* barrier means they already share the same orbit representative without any alignment (Figure 1, Consequence panel).

This paper introduces THEORY-INIT, an initialization that selects a canonical orbit representative at $t{=}0$: instead of letting the optimizer pick a random arrangement of neurons, we place known features into fixed neuron slots and keep them there through depth, so two runs land in the same arrangement rather than two random ones (Figure 1). The construction has three pieces. The first layer is built from a constructive approximation of the target (Yarotsky, 2017; Elbrächter et al., 2021), binding specific neuron indices to specific functional roles. Identity-plus-noise middle layers ($I{+}\Delta$, $\sigma_\Delta \ll 1$) then preserve that assignment through depth, both at the level of activations, where $\mathbb{E}\|h_\ell - h_1\|^2 \leq [(1{+}w\sigma^2)^{\ell-1}{-}1]\|h_1\|^2$ holds unconditionally for the actual ReLU recursion when $h_1 \geq 0$, and at the level of gradients, through a near-isometric middle-block Jacobian. A small-random output layer learns how to combine the resulting features. We verify the construction[1] on four regression tasks (UCI Yacht, Energy, Concrete, MoleculeNet Lipophilicity): two THEORY-INIT networks trained from different seeds end up an order of magnitude closer in weight space than two He networks, and Git Re-Basin matching returns the identity permutation. The same behaviour replicates on a small CNN at MNIST scale, and on CIFAR-10 once the prescribed filters cover all input channels. The source of this symmetry-breaking behavior is the construction itself (fixed first-layer placement carried through depth by $I{+}\Delta$ middle

[1]Code: https://github.com/RandomAnass/theory-init.

layers), and the choice of theory contributes accuracy, not proximity, and only when the theory rows align with the dimensions the trained network uses.

2. Symmetry-breaking initialization

Setup. We consider feedforward ReLU MLPs $f_\theta(x) = W_L\,\mathrm{ReLU}(W_{L-1}\cdots\mathrm{ReLU}(W_1x+b_1)\cdots)+b_L$ with architecture $[d_{\text{in}}, w, w, w, 1]$, w=64, L=4. Given an approximation $\hat{f}$ of the target (Taylor or Chebyshev coefficients, or an established domain theory such as Crippen atomic contributions for molecular LogP (Wildman & Crippen, 1999), Froude scaling for hull resistance, or an Abrams cement-water law), THEORY-INIT constructs θ_0 in three stages (Figure 1; Algorithm 1). The construction reuses standard piecewise-linear (PWL) approximations of $\hat{f}$ from constructive-approximation theory (Yarotsky, 2017; Elbrächter et al., 2021). Where a closed-form or smooth target is available, the appendix collects an explicit *construction library* drawn from that line of work (Section C.7), an alternative Stage-1 source to the per-dataset prescriptions in Section B.2, with sup-norm error bounds carried over verbatim from the cited papers.

Construction. We encode $\hat{f}$ on $[0,1]$ via the standard PWL hinge representation $\hat{f}_{\text{pwl}}(z) = v_0 + m_0 z + \sum_{i=1}^{N}(m_i - m_{i-1})\,\mathrm{ReLU}(z - t_i)$ at N interior breakpoints $\{t_1,\ldots,t_N\}$ (one hinge per interior knot), and write each hinge $\mathrm{ReLU}(u_i^\top x - t_i)$ into a fixed first-layer slot $W_{1,i,:}{=}u_i^\top$, $b_{1,i}{=}-t_i$ (i=1, …, N). We call these N deterministic rows the *theory rows* (their convolutional counterpart *theory filters*, used in Section D.7). The remaining $w-N$ *padding rows* are $\mathcal{N}(0, \alpha_1^2 \cdot 2/d_{\text{in}})$ at small scale ($\alpha_1 \le 0.1$). Each theory-row index is bound to a specific role (breakpoint t_i); this fixes a canonical orbit representative of the layer-1 placement and leaves only the padding rows exchangeable (stabilizer S_{w-N}, residual orbit size $w!/(w-N)!$). Middle layers ℓ=2, …, $L-1$ use

$$W_\ell = I_w + \Delta_\ell, \quad b_\ell{=}0, \quad (\Delta_\ell)_{ij} \sim \mathcal{N}(0, \sigma_\Delta^2), \tag{1}$$

with $\sigma_\Delta \in \{0, 10^{-3}, 10^{-2}\}$ (*dead*, *alive*, *alive-high*, chosen by data regime). The output layer $W_L \sim \mathcal{N}(0, \alpha_{\text{out}}^2 \cdot 2/w)$ with α_{out}=0.1 is small-random; training learns to combine the theory-feature responses.

3. Signal Preservation

The identity-plus-noise construction avoids a known obstacle: i.i.d. Gaussian initialization cannot achieve dynamical isometry (singular values of the Jacobian concentrating at 1) because the Jacobian's singular spectrum spreads with depth rather than concentrating: even at critical tuning its variance grows linearly in depth (Pennington et al., 2017).

Algorithm 1 THEORY-INIT initialization

Require: arch $[d_{\text{in}}, w, \ldots, w, 1]$, theory $\hat{f}$, noise σ_Δ
1: Pick N breakpoints $\{t_i\}$; compute $v_i = \hat{f}(t_i)$ and slope changes $m_i - m_{i-1}$
2: $W_1 \leftarrow [W_{\text{theory}}; W_{\text{He-small}}]$, $b_1 \leftarrow [b_{\text{theory}}; 0]$ (§B.2)
3: **for** $\ell = 2$ **to** $L-1$ **do**
4: $W_\ell \leftarrow I_w + \Delta_\ell$, $b_\ell \leftarrow 0$ (Eq. 1)
5: **end for**
6: $W_L \sim \mathcal{N}(0,\ \alpha_{\text{out}}^2 \cdot 2/w)$ with $\alpha_{\text{out}} = 0.1$, $b_L = 0$
7: **return** θ_0

Starting from a near-identity matrix $W_\ell = I + \Delta_\ell$ (cf. Pan et al., 2025) instead of an i.i.d. matrix gives a near-isometric Jacobian directly, with explicit, depth-dependent control:

Proposition 3.1 (Linearized signal preservation through $I + \Delta$ layers)**.** *Let $h_1 \in \mathbb{R}^w$ be a fixed first-layer activation, and define the linearized recursion $\tilde{h}_{k+1} = (I_w + \Delta_k)\,\tilde{h}_k$ for $k = 1, \ldots, \ell-1$ with $(\Delta_k)_{ij} \sim \mathcal{N}(0, \sigma^2)$ i.i.d. across k, i, j, $\tilde{h}_1 = h_1$. Then*

$$\mathbb{E}\Big[\|\tilde{h}_\ell - h_1\|^2\Big] = \big[(1 + w\sigma^2)^{\ell-1} - 1\big]\,\|h_1\|^2. \tag{2}$$

The actual ReLU recursion $h_{k+1} = \mathrm{ReLU}((I_w + \Delta_k)\,h_k)$ coincides with $\tilde{h}_k$ pathwise on the no-clipping event $\mathcal{T}_\ell := \{(I_w{+}\Delta_k)\tilde{h}_k \ge 0$ coordinatewise for all $k \le \ell-1\}$ (so (2) also describes the actual activations on $\mathcal{T}_\ell$). For $w = 64$, $\sigma = 0.01$, $\ell - 1 = 2$ (our default real-data architecture), the squared deviation $\mathbb{E}\|\tilde{h}_\ell - h_1\|^2/\|h_1\|^2$ is at most 1.28%, i.e. relative L2 deviation $\le 11.3\%$.

Proposition 3.2 (Unconditional L^2-mean bound for the actual ReLU recursion)**.** *The right-hand side of (2) is also an unconditional upper bound for the actual ReLU recursion $h_{k+1} = \mathrm{ReLU}((I_w + \Delta_k)h_k)$ when $h_1 \ge 0$ coordinatewise (automatic, as h_1 is itself a ReLU output): 1-Lipschitzness of ReLU and $\mathrm{ReLU}(h_1) = h_1$ give $\|h_{k+1}{-}h_1\| \le \|(h_k{-}h_1){+}\Delta_k h_k\|$, and conditioning on $\Delta_{<k}$ collapses the cross term, yielding $\mathbb{E}\|h_\ell{-}h_1\|^2 \le [(1{+}w\sigma^2)^{\ell-1}{-}1]\,\|h_1\|^2$ without any no-clipping hypothesis (Jacobian near-isometry in Corollary 3.3 still uses $\mathcal{T}_\ell$).*

Corollary 3.3 (Near-isometric middle-block Jacobian)**.** *Let $J_{\text{mid}} := \prod_{k=1}^{\ell-1}(I_w + \Delta_k)$ be the linearized middle-block Jacobian. With probability at least $1 - 2(\ell-1)\exp(-w/2)$, every singular value of J_{mid} lies in $[(1 - 3\sigma\sqrt{w})^{\ell-1},\ (1 + 3\sigma\sqrt{w})^{\ell-1}]$. On the no-clipping (transparency) event $\mathcal{T}_\ell$ this is also the Jacobian of the actual ReLU middle block; the empirical $\mathcal{T}_\ell$-rate on the deployed architecture is only* 49–80% *(Theorem A.9), so this actual-ReLU clause is* approximate, not nearly-sure *in our experiments. The unconditional L^2-mean preservation of Theorem 3.2 (which holds with no transparency hypothesis) is what carries the index-preservation argument below. The same event also gives $\|J_{\text{mid}} - I_w\|_{\text{op}} \le (1 + 3\sigma\sqrt{w})^{\ell-1} - 1$, so J_{mid}*

is close to the identity, not merely norm-preserving. For $w = 64, \sigma = 0.01, \ell - 1 = 2$ the singular values lie in $[0.578, 1.538]$ with probability $\geq 1 - 4e^{-32}$, but at this σ the op-norm-to-identity bound (≈ 0.538) is loose; index preservation here rests on the explicit $I + \Delta$ structure of each factor rather than on the singular-value interval alone (tight at $\sigma = 10^{-3}$, where the bound is ≈ 0.049; Theorem A.14).

Index assignment is preserved through depth. Together these say the hidden representation evolves only weakly through Stage 3, so the canonical neuron-index assignment imposed by Stage 2 carries through to the output layer: theory row i at layer 1 routes (up to small noise) to coordinate i at every subsequent layer. A random matrix would scramble this routing; $I + \Delta$ does not. The construction therefore places θ_0 at a specific representative of the permutation orbit rather than at a uniformly-random one: meaningful features sit in fixed slots, near-identity middle layers keep slot i at slot i, and the output layer learns how to combine them. Section 4 measures the empirical consequence on cross-seed weight-space proximity; full proofs are in the appendix.

4. Empirical results

Setup. We evaluate on four regression tasks: **Yacht Hydrodynamics** (UCI, n=308, d=6), **Energy Efficiency** (UCI, n=768, d=8), **Concrete Strength** (UCI, n=1,030, d=8), and **MoleculeNet Lipophilicity** (LogP from atom-type counts, n=4,200, d=64). Table 1 reports the LMC results across all four. Architecture is $[d_{\text{in}}, 64, 64, 64, 1]$ for the MLPs (the CNN setup is in Section D.7). We compare THEORY-INIT against He (He et al., 2015) and orthogonal (Saxe et al., 2014) initialization, with 4 seeds per (dataset, init). For LMC, we use a fixed train/test split so all paired networks are trained and evaluated on the same data; only the init seed varies.

Held-out accuracy. THEORY-INIT achieves lower MSE than He on UCI Yacht and Energy (medians of 0.002 vs. 0.024 and 0.002 vs. 0.013 over 20 seeds), and a smaller but consistent gap on LogP (0.37 vs. 0.44). The improvement is not specific to He: against a nine-method baseline (He, Xavier, orthogonal, LeCun, LSUV, ZerO, MetaInit, GradInit, data-dependent; LeCun et al., 1998; Mishkin & Matas, 2016; Dauphin & Schoenholz, 2019; Zhu et al., 2021), THEORY-INIT places at mean rank $1.71/9$ across the five UCI regression datasets where a physics prior is available (Table 8).

Initialization versus feature, residual, frozen on Lipophilicity. On Lipophilicity we compare THEORY-INIT against three controls that share the same Crippen knowledge but deliver it differently (residual, feature augmentation, and frozen layer-1 theory; full forms in Section D.2). Across 20 seeds, all three land within $\pm 1.3\%$ of He, while THEORY-INIT achieves a -7.0% mean (-8.2% median) RMSE reduction: the gain comes from delivering the prescription as an initialization, not from the choice of theory. A random-Gaussian-with-matched-magnitude first-layer control (Section G.1) reproduces the symmetry-breaking signature on all four datasets without any Crippen content.

Pre-alignment in weight space. If Stage 2 fixes a canonical row placement and Stage 3 preserves it through depth (Proposition 3.1 and Corollary 3.3), then THEORY-INIT nets trained from different init seeds should occupy a structurally similar region of weight space, and Git Re-Basin (Ainsworth et al., 2023) permutation matching should not find a nontrivial alignment.

For each (dataset, init) we train four nets and form all $\binom{4}{2} = 6$ within-scheme pairs. For each pair (θ_A, θ_B) we measure the naive barrier $\mathcal{B}_{\text{nv}}(\theta_A, \theta_B) = \max_\alpha \mathcal{L}((1-\alpha)\theta_A + \alpha\theta_B) - \frac{1}{2}[\mathcal{L}(\theta_A) + \mathcal{L}(\theta_B)]$ over $\alpha \in [0, 1]$ with $\mathcal{L}$ the test MSE, the aligned barrier $\mathcal{B}_{\text{al}} = \mathcal{B}_{\text{nv}}(\theta_A, P^*\theta_B)$ under the layer-wise permutation P^* found by Ainsworth-style weight matching, and the post-alignment distance $d = \|\theta_A - P^*\theta_B\|_F$. The init-time distance $d_0 = \|\theta_0^{(A)} - \theta_0^{(B)}\|_F$ is at $t = 0$.

Figure 2 and Table 1 summarize the findings; Figure 3 (appendix) shows the trained pairs sitting in a visibly flatter, narrower sub-basin under THEORY-INIT than under He or orthogonal init in the filter-normalized 2D landscape.

The scaffold, not the theory content, drives the proximity, and it rescales with the prescribed fraction. Two THEORY-INIT networks initialized with different seeds start an order of magnitude closer in Frobenius distance than two He networks on the three UCI datasets (d_0=2 vs. 28), and several times closer on Lipophilicity, where LogP uses many more padding rows than theory rows (Table 1). This gap survives a scale-free comparison: dividing the trained-time distance by each pair's weight norm leaves THEORY-INIT at $\hat{d}_{\text{nv}} = 0.54$–$1.04$ versus He's 1.40–1.44 (Table 10), so it is not an artifact of THEORY-INIT's smaller weight mass. A breadth sweep over four depths and widths confirms the signature is not specific to this architecture: after scale normalization THEORY-INIT sits at $\hat{d}_0$=0.18 versus He/Ortho at the random baseline ≈ 1.41, and the identity-matching property *strengthens* with depth (up to $22/24$ pairs in five-hidden-layer nets; Section D.8). The mechanism is Stage 2: it selects a single canonical placement of the theory rows, leaving only the padding rows random across seeds. Two controls show this proximity comes from the deterministic, low-magnitude *scaffold* rather than from the specific theory content. (i) A tiny-He baseline (β=0.1, Section D.6) reaches d_0=2.8 on Yacht

Data	Pair	$\mathcal{B}_{\text{nv}}$	$\mathcal{B}_{\text{al}}$	d_0	d_{nv}	d_{al}
Yacht	He–He	1.20	0.43	27.8	28.8	22.1
	Ortho–Ortho	1.03	0.11	16.3	18.2	14.8
	THEORY-INIT–THEORY-INIT	**0.29**	**0.20**	**2.0**	**8.1**	**8.0**
Energy	He–He	0.84	0.63	27.9	29.2	22.9
	Ortho–Ortho	0.80	0.33	16.5	18.5	15.5
	THEORY-INIT–THEORY-INIT	**0.55**	**0.51**	**2.0**	**10.3**	**10.2**
Concrete	He–He	1.11	0.82	27.9	31.6	25.0
	Ortho–Ortho	0.82	0.44	16.5	22.3	17.2
	THEORY-INIT–THEORY-INIT	**0.61**	**0.48**	**2.1**	**13.9**	**13.9**
LogP	He–He	0.81	1.04	27.7	29.9	25.8
	Ortho–Ortho	0.53	0.69	19.5	24.4	19.8
	THEORY-INIT–THEORY-INIT	**0.48**	**0.41**	**4.9**	**16.4**	**15.9**

Table 1. LMC summary (medians over 6 within-scheme pairs; full per-pair records in Table 10). d_0 = init-time distance, $d_{\text{nv}}, d_{\text{al}}$ = trained-time distance before/after Git Re-Basin alignment.

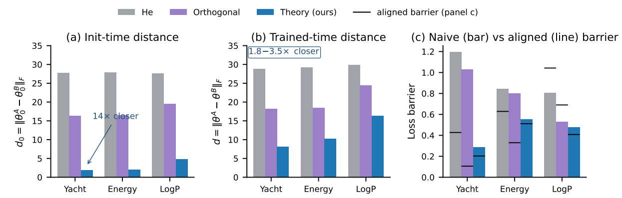

Figure 2. **THEORY-INIT seeds share an orbit representative.** Medians over 6 within-scheme pairs per dataset; lower is better. **(a)** Init-time distance $d_0 = \|\theta_0^A - \theta_0^B\|_F$, 13–14× smaller for THEORY-INIT than He on UCI tasks ($\sim 6\times$ on LogP). **(b)** Trained-time naive distance; the gap shrinks but persists. **(c)** Naive (bar) vs. Git Re-Basin aligned (line) loss barriers.

versus THEORY-INIT's 1.96, so the low first-layer magnitude accounts for most of the reduction and deterministic placement contributes the residual $1.4\times$. (ii) Replacing the theory rows with random Gaussian rows of matched magnitude reproduces the same proximity, identity-matching, and barrier reduction on all four datasets (Section G.1), so any deterministic magnitude-matched scaffold suffices; the theory content adds accuracy, not proximity. The same asymmetry rescales on a small CNN with prescribed theory filters in the first conv layer: it persists at MNIST scale and on CIFAR-10 once the theory filters cover all input channels, with the size of the effect tracking the prescribed fraction of the first layer (Section D.7 and Table 14). Strict $P^*{=}I$ is small-MLP only ($0/6$ pairs at both CNN settings).

Git Re-Basin returns the identity on THEORY-INIT–THEORY-INIT pairs. On THEORY-INIT pairs, weight matching leaves the post-alignment distance essentially unchanged: $d_{\text{al}}/d_{\text{nv}}$ is within a percent or two on all four datasets (medians 0.97–0.999), so the matched network is already in the same neuron frame. The strict identity permutation is recovered in 3–5 of 6 pairs per dataset at the deployed $\sigma_\Delta{=}10^{-2}$ (Section D.6); it strengthens with depth, reaching $22/24$ in the five-hidden-layer breadth configurations (Table 17). The *barriers* need not move by the same tiny amount, because a barrier is a maximum of the loss over the interpolation path, not a function of the Frobenius distance: aligning θ_B to a near-identical frame still changes the path, so the aligned barrier can drop modestly even when the distance is unchanged (per-dataset drops in Table 1). He pairs differ: alignment cuts barrier and distance more on Yacht and Energy and can even raise the LogP barrier slightly (Table 1).

Which barrier matters: THEORY-INIT is lowest on naive, orthogonal lowest on aligned. THEORY-INIT has the lowest *naive* barrier on all four datasets, the metric matched to our claim: it measures how close two independently trained nets are *before* any alignment search. The *aligned* barrier ordering differs: it is higher for THEORY-INIT than for Ortho on three of four datasets (Yacht $0.20 > 0.11$, Energy $0.51 > 0.33$, Concrete $0.48 > 0.44$; LogP is the exception, $0.41 < 0.69$). This is expected rather than contradictory. Orthogonal init keeps the full permutation symmetry, so the matching search has a large barrier to remove and reaches a low aligned floor; THEORY-INIT removed that symmetry by construction, so there is little left to remove and alignment barely moves the barrier. The aligned barrier therefore chiefly measures the residual exploitable symmetry a scheme left behind, which THEORY-INIT gives up on purpose. The identity result matches Proposition 3.1 together with the layer-1 trace-dominance argument (Lemma A.19, Corollary A.21) at $t{=}0$; persistence at trained time is an empirical extension. To check that the identity is *placement* rather than a numerical accident of the cost matrix, we randomly permute the theory rows of θ_B's layer-1 and re-run matching: LSAP recovers the inverse scramble (Section D.4).

Discussion. Strict LMC does not hold on these small MLPs under any initialization: naive barriers exceed the trained test MSE for every scheme. What we report is the *relative* asymmetry of naive barriers across schemes; the symmetry-breaking signature is scaffold-driven, not theory-correctness-driven, and the appendix isolates this from train-MSE, magnitude, and CNN confounds (Sections D.6 and G.1). The construction targets neuron-permutation symmetry only; continuous gauges (ReLU positive-scaling, attention rotational gauge) and padding-row exchangeability remain (Sections E.2 and E.3).

References

M. Adnan, R. Jain, E. Sharma, R. G. Krishnan, Y. Ioannou. Sparse Training from Random Initialization: Aligning Lottery Ticket Masks using Weight Symmetry. *ICML*, 2025.

S. K. Ainsworth, J. Hayase, S. Srinivasa. Git Re-Basin: Merging Models modulo Permutation Symmetries. *ICLR*, 2023.

R. Burkard, M. Dell'Amico, S. Martello. *Assignment Problems*. SIAM, 2009.

W. M. Czarnecki, S. Osindero, M. Jaderberg, G. Świrszcz, R. Pascanu. Sobolev Training for Neural Networks. *NeurIPS*, 2017.

Y. N. Dauphin, S. Schoenholz. MetaInit: Initializing learning by learning to initialize. *NeurIPS*, 2019.

K. R. Davidson, S. J. Szarek. Local operator theory, random matrices and Banach spaces. In *Handbook of the Geometry of Banach Spaces*, Vol. I, pp. 317–366. North-Holland, 2001.

F. Draxler, K. Veschgini, M. Salmhofer, F. A. Hamprecht. Essentially No Barriers in Neural Network Energy Landscape. *ICML*, 2018.

D. Elbrächter, D. Perekrestenko, P. Grohs, H. Bölcskei. Deep Neural Network Approximation Theory. *IEEE Transactions on Information Theory*, 67(5):2581–2623, 2021.

R. Entezari, H. Sedghi, O. Saukh, B. Neyshabur. The Role of Permutation Invariance in Linear Mode Connectivity of Neural Networks. *ICLR*, 2022.

J. Frankle, M. Carbin. The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks. *ICLR*, 2019.

J. Frankle, G. K. Dziugaite, D. Roy, M. Carbin. Linear Mode Connectivity and the Lottery Ticket Hypothesis. *ICML*, 2020.

T. Garipov, P. Izmailov, D. Podoprikhin, D. P. Vetrov, A. G. Wilson. Loss Surfaces, Mode Connectivity, and Fast Ensembling of DNNs. *NeurIPS*, 2018.

X. Glorot, Y. Bengio. Understanding the difficulty of training deep feedforward neural networks. *AISTATS*, 2010.

K. He, X. Zhang, S. Ren, J. Sun. Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification. *ICCV*, 2015.

K. Jordan, H. Sedghi, O. Saukh, R. Entezari, B. Neyshabur. REPAIR: REnormalizing Permuted Activations for Interpolation Repair. *ICLR*, 2023.

Y. LeCun, L. Bottou, G. B. Orr, K.-R. Müller. Efficient BackProp. In *Neural Networks: Tricks of the Trade*, Springer LNCS 1524, pp. 9–50, 1998.

H. Li, Z. Xu, G. Taylor, C. Studer, T. Goldstein. Visualizing the Loss Landscape of Neural Nets. *NeurIPS*, 2018.

D. Lim, T. Putterman, R. Walters, H. Maron, S. Jegelka. The Empirical Impact of Neural Parameter Symmetries, or Lack Thereof. *NeurIPS*, 2024.

D. Mishkin, J. Matas. All you need is a good init. *ICLR*, 2016.

I. Sadrtdinov, D. Pozdeev, D. Vetrov, E. Lobacheva. To Stay or Not to Stay in the Pre-train Basin: Insights on Ensembling in Transfer Learning. *NeurIPS*, 2023.

R. Hecht-Nielsen. On the algebraic structure of feedforward network weight spaces. In R. Eckmiller (ed.), *Advanced Neural Computers*, North-Holland, pp. 129–135, 1990.

Y. Pan, C. Wang, Z. Wu, Q. Wang, M. Zhang, Z. Xu. IDInit: A Universal and Stable Initialization Method for Neural Network Training. *ICLR*, 2025.

J. Pennington, S. Schoenholz, S. Ganguli. Resurrecting the sigmoid in deep learning through dynamical isometry: theory and practice. *NeurIPS*, 2017.

M. Raissi, P. Perdikaris, G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving non-linear partial differential equations. *J. Comp. Physics*, 378:686–707, 2019.

A. M. Saxe, J. L. McClelland, S. Ganguli. Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. *ICLR*, 2014.

S. P. Singh, M. Jaggi. Model Fusion via Optimal Transport. *NeurIPS*, 2020.

G. Stoica, D. Bolya, J. Bjorner, P. Ramesh, T. Hearn, J. Hoffman. ZipIt! Merging Models from Different Tasks without Training. *ICLR*, 2024.

A. Theus, A. Cabodi, S. Anagnostidis, A. Orvieto, S. P. Singh, V. Boeva. Generalized Linear Mode Connectivity for Transformers. *NeurIPS*, 2025.

R. Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge University Press, 2018.

M. Wortsman, G. Ilharco, S. Y. Gadre, R. Roelofs, R. Gontijo-Lopes, A. S. Morcos, H. Namkoong, A. Farhadi, Y. Carmon, S. Kornblith, L. Schmidt. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. *ICML*, 2022.

S. A. Wildman, G. M. Crippen. Prediction of Physicochemical Parameters by Atomic Contributions. *J. Chem. Inf. Comput. Sci.*, 39:868–873, 1999.

D. Yarotsky. Error bounds for approximations with deep ReLU networks. *Neural Networks*, 94:103–114, 2017.

J. Zhao, F. T. Schaefer, A. Anandkumar. ZerO Initialization: Initializing Neural Networks with only Zeros and Ones. *TMLR*, 2022.

C. Zhu, R. Ni, Z. Xu, K. Kong, W. R. Huang, T. Goldstein. GradInit: Learning to initialize neural networks for stable and efficient training. *NeurIPS*, 2021.

Appendix

Contents.

- §A Theory: full proofs and remarks (signal preservation, ReLU transparency, near-isometric Jacobian, cost-matrix diagonal dominance, identity-Hungarian-optimality)
- §B Method details: PWL physics-neuron construction and per-dataset hyperparameters
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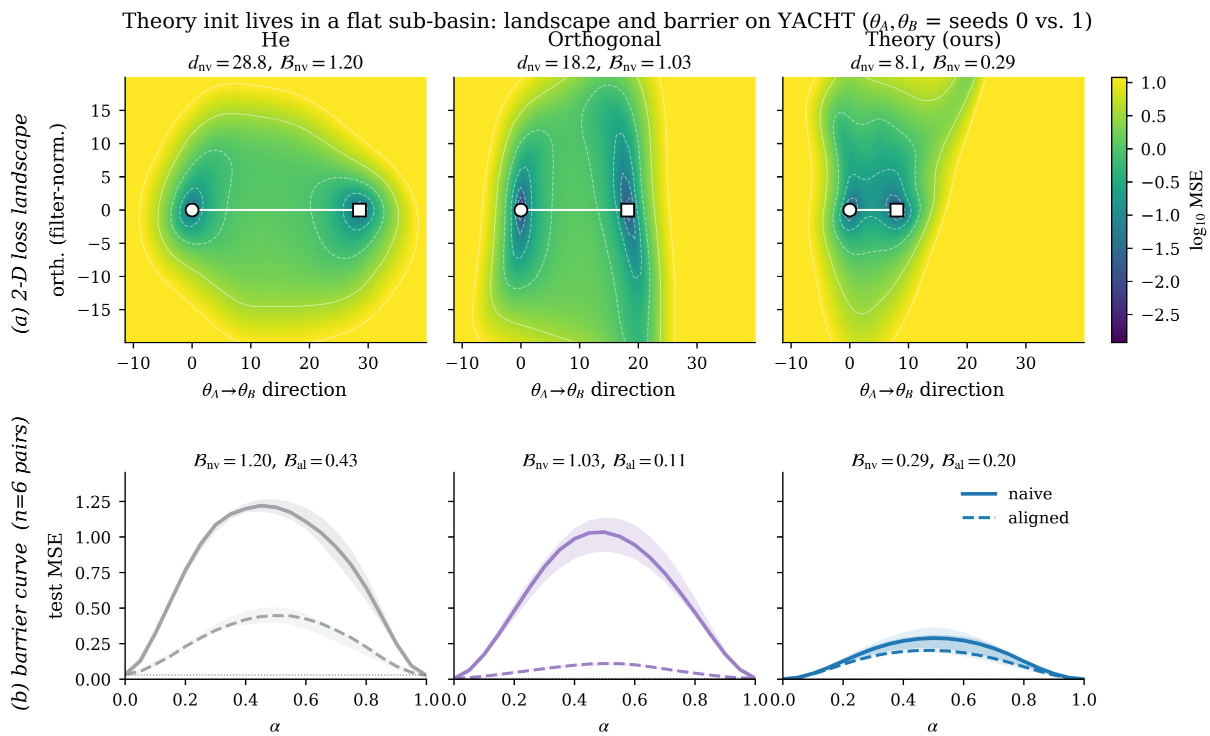

Figure 3. **THEORY-INIT occupies a flatter, narrower sub-basin between trained pairs.** Two seeds of THEORY-INIT, He, and orthogonal init are trained on Yacht; for each pair we plot the loss on the 2D plane spanned by the interpolation direction $\theta_A \to \theta_B$ and a random orthogonal direction (filter-normalized; Li et al. (2018)). **(a)** 2D log-MSE landscape with circle/square markers at θ_A, θ_B. Theory pairs sit much closer (d_{nv}=8.1 vs. 28.8 for He) and the basin connecting them is shallower. **(b)** Loss along the linear interpolation $\alpha \in [0, 1]$, naive (solid) vs. Git Re-Basin aligned (dashed); median $\pm$ IQR over 6 within-scheme pairs. THEORY-INIT already starts close to the aligned curve, while alignment flattens the He and Orthogonal barriers.

A. Full proofs and remarks

This appendix supplies the full proofs of the theoretical claims of the main paper. Proposition A.2 (§A.2) is the signal-preservation theorem (Prop. 3.1 of the main paper). Proposition A.5 (§A.3) lifts the linearized identity to an unconditional L^2-mean upper bound for the actual ReLU recursion when $h_1 \geq 0$ coordinatewise (the regime in which THEORY-INIT operates by construction). Proposition A.7 (§A.4) gives a sufficient no-clipping probability bound under the strictly-positive activation precondition $\min_j h_{1,j} \geq c > 0$ (still required for the pathwise Jacobian lift, but no longer for the L^2-mean preservation in Theorem A.5). Corollary A.12 (§A.5) is the near-isometric middle-block Jacobian (Cor. 3.3 of the main

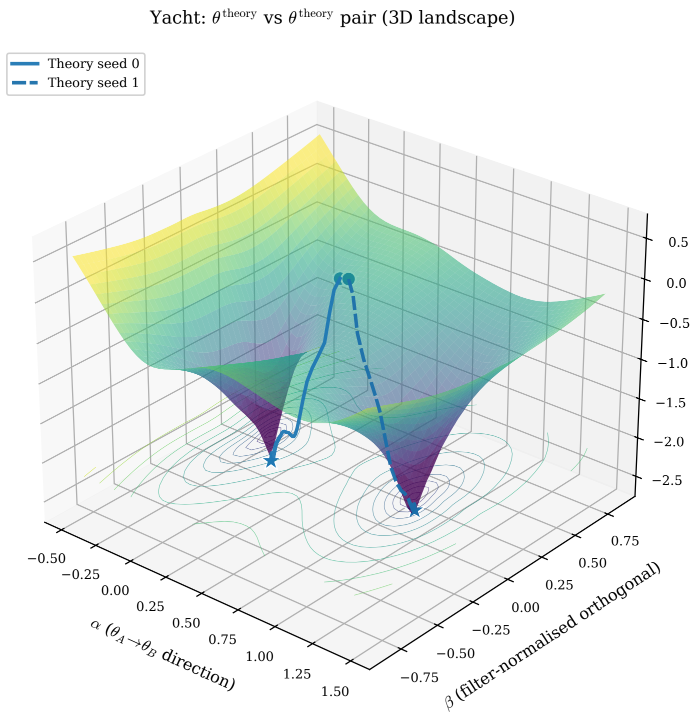

Figure 4. 3D companion view of Figure 3: $\log_{10}$ MSE surface for two THEORY-INIT seeds trained on Yacht (seeds 0 and 1), 51×51 grid in the ($\theta_A \to \theta_B$, filter-normalized orthogonal) plane. Init markers (open circles) sit on the upper plateau; final markers (stars) at the bottom of two distinct minima. Trajectories begin close together and diverge to opposite minima along the interpolation axis.

paper). Lemma A.19 and Corollary A.21 (§A.8) prove that the Hungarian-optimal layer-1 permutation is the identity for theory–theory pairs, predicting the empirical Git Re-Basin result on THEORY-INIT–THEORY-INIT pairs.

A.1. Setting and notation

The network is $f_\theta(x) = W_L\,\mathrm{ReLU}(W_{L-1}\cdots\mathrm{ReLU}(W_1x+b_1)\cdots)+b_L$ with hidden width w and $L=4$ (three hidden layers). Post-ReLU activations are h_ℓ, with $h_0 = x$, $h_\ell = \mathrm{ReLU}(W_\ell h_{\ell-1}+b_\ell)$. The first layer W_1 contains N *physics neurons* computing prescribed PWL features $\mathrm{ReLU}(x\cdot u - t_i)$ (input direction u, breakpoint t_i), plus $w-N$ small-random padding neurons. Middle layers $\ell = 2,\ldots,L-1$ are

$$W_\ell = I_w + \Delta_\ell, \quad b_\ell = 0, \quad (\Delta_\ell)_{ij} \overset{\text{iid}}{\sim} \mathcal{N}(0,\sigma^2),$$

with Δ_ℓ jointly independent across ℓ. The output layer is $W_L \sim \mathcal{N}(0, 0.01\cdot 2/w)$. We work at initialization (no training has occurred) and condition on a fixed realized first-layer activation $h_1 = h_1(x)$. The theory block of h_1 is deterministic given x, while the padding block is random through W_P; the linearized signal-preservation identity holds for every fixed h_1.

Key abbreviation. For $\ell \geq 2$, $W_\ell h_{\ell-1} + b_\ell = h_{\ell-1} + \Delta_\ell h_{\ell-1}$, and ReLU is *transparent* at layer ℓ if $h_{\ell-1} + \Delta_\ell h_{\ell-1} \geq 0$ element-wise. Independence: Δ_ℓ is independent of everything that has happened before layer ℓ.

Assumption A.1 (ReLU transparency). For all $k = 1,\ldots,\ell-1$, $h_k + \Delta_k h_k \geq 0$ element-wise.

Strictly-positive illustrative regime. $w = 64$, $L = 4$ so $\ell - 1 = 2$ middle hops, $\sigma \in \{0, 10^{-3}, 10^{-2}\}$. For numerical illustrations of the optional no-clipping bound (Theorem A.7) we sometimes assume $c = \min_j (h_1)_j \geq 0.1\,\|h_1\|$; this is *not* a property the raw PWL+padding architecture satisfies (Theorem A.8), and the L^2-mean preservation (Theorem A.5) does not require it.

A.2. Proof of Theorem A.2 (signal preservation)

Proposition A.2 (Linearized signal preservation)**.** *Let $h_1 \in \mathbb{R}^w$ be a fixed first-layer activation, and define the* linearized *recursion*

$$\tilde h_{k+1} = (I_w + \Delta_k)\,\tilde h_k \qquad (k = 1,\ldots,\ell-1), \qquad \tilde h_1 = h_1, \tag{3}$$

with $(\Delta_k)_{ij} \overset{iid}{\sim} \mathcal{N}(0,\sigma^2)$ jointly independent across k. Then

$$\mathbb{E}\big[\|\tilde h_\ell - h_1\|^2\big] = \big[(1+w\sigma^2)^{\ell-1} - 1\big]\cdot\|h_1\|^2. \tag{4}$$

This is an unconditional *identity for the linearized recursion (no transparency hypothesis is used). On the no-clipping event $\mathcal{T}_\ell := \{(I_w + \Delta_k)\tilde{h}_k \geq 0$ coordinatewise for all $k \leq \ell - 1\}$, the actual ReLU recursion $h_{k+1} = \mathrm{ReLU}((I_w + \Delta_k)h_k)$ coincides with $\tilde{h}_k$ pathwise (Theorem A.3), so (4) also describes the actual ReLU activations on $\mathcal{T}_\ell$.*

Proof. Define $e_k := \tilde{h}_{k+1} - \tilde{h}_k = \Delta_k \tilde{h}_k$. Telescoping,

$$\tilde{h}_\ell - h_1 = \sum_{k=1}^{\ell-1} e_k,$$

$$\|\tilde{h}_\ell - h_1\|^2 = \sum_{k=1}^{\ell-1} \|e_k\|^2 + 2 \sum_{1 \leq j < k \leq \ell-1} \langle e_j, e_k \rangle.$$

Diagonal terms. Conditional on $\tilde{h}_k$, coordinate i of $\Delta_k \tilde{h}_k$ is $\sum_{j=1}^{w} (\Delta_k)_{ij} (\tilde{h}_k)_j \sim \mathcal{N}(0, \sigma^2 \|\tilde{h}_k\|^2)$, so

$$\mathbb{E}[\|e_k\|^2 \mid \tilde{h}_k] = w\sigma^2 \, \|\tilde{h}_k\|^2, \quad \mathbb{E}[\|e_k\|^2] = w\sigma^2 \, \mathbb{E}[\|\tilde{h}_k\|^2].$$

Cross terms. For $j < k$, $\tilde{h}_k$ and e_j depend only on $\Delta_1, \dots, \Delta_{k-1}$, while Δ_k is independent of these. Conditioning on $(\tilde{h}_k, e_j)$,

$$\mathbb{E}[\langle e_j, \Delta_k \tilde{h}_k \rangle \mid \tilde{h}_k, e_j] = \langle e_j, \mathbb{E}[\Delta_k] \, \tilde{h}_k \rangle = 0.$$

Norm recursion. $\|\tilde{h}_{k+1}\|^2 = \|\tilde{h}_k\|^2 + 2\langle \tilde{h}_k, \Delta_k \tilde{h}_k \rangle + \|\Delta_k \tilde{h}_k\|^2$ gives $\mathbb{E}[\|\tilde{h}_{k+1}\|^2 \mid \tilde{h}_k] = (1 + w\sigma^2) \, \|\tilde{h}_k\|^2$, hence $\mathbb{E}[\|\tilde{h}_k\|^2] = (1 + w\sigma^2)^{k-1} \, \|h_1\|^2$.

Assembly. $\mathbb{E}[\|\tilde{h}_\ell - h_1\|^2] = \sum_{k=1}^{\ell-1} w\sigma^2 (1 + w\sigma^2)^{k-1} \|h_1\|^2 = [(1 + w\sigma^2)^{\ell-1} - 1] \, \|h_1\|^2$. ☐

Lemma A.3 (Linearized–ReLU coupling)**.** *With $\tilde{h}_k$ as in (3) and $h_{k+1} := \mathrm{ReLU}((I_w + \Delta_k) \, h_k)$ starting from $h_1 = \tilde{h}_1$, the two recursions agree pathwise on the no-clipping event $\mathcal{T}_\ell$: $h_k = \tilde{h}_k$ for all $k \leq \ell$.*

Proof. Induction on k. Base: $h_1 = \tilde{h}_1$. Step: if $h_k = \tilde{h}_k$ and $(I_w + \Delta_k)\tilde{h}_k \geq 0$ coordinatewise, then $\mathrm{ReLU}((I_w + \Delta_k)h_k) = (I_w + \Delta_k)\tilde{h}_k = \tilde{h}_{k+1}$. ☐

Remark A.4 (Numerical implications)*.* For $w = 64$, $L = 4$, $\ell - 1 = 2$, the relative L2 deviation $\sqrt{(1 + w\sigma^2)^{\ell-1} - 1}$ is 0% at $\sigma{=}0$ (exact identity), 1.13% at $\sigma{=}10^{-3}$, and 11.33% at $\sigma{=}10^{-2}$. The squared fraction $\mathbb{E}\|h_3 - h_1\|^2 / \|h_1\|^2$ at $\sigma{=}10^{-2}$ is $\approx 1.28\%$ (which is what the main-paper text reports as "deviation").

A.3. Unconditional actual-ReLU preservation

Theorem A.2 is exact for the linearized recursion and pathwise sharp on the no-clipping event $\mathcal{T}_\ell$. The next proposition removes the no-clipping precondition: the same right-hand side is also an unconditional L^2-mean upper bound for the actual ReLU recursion whenever h_1 is non-negative coordinatewise. In particular, this regime covers THEORY-INIT by construction, since h_1 is itself a ReLU output, hence componentwise non-negative.

Proposition A.5 (Unconditional actual-ReLU signal preservation)**.** *Let* ReLU *act coordinatewise. Fix a non-negative first-layer activation $h_1 \in \mathbb{R}^w_{\geq 0}$ and define the actual middle-block recursion*

$$h_{k+1} \;=\; \mathrm{ReLU}\big((I_w + \Delta_k) \, h_k\big), \qquad k = 1, \dots, \ell - 1,$$

with $(\Delta_k)_{ij} \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$ jointly independent across k. Then for every $\ell \geq 1$,

$$\mathbb{E}\big[\|h_\ell\|^2\big] \;\leq\; (1 + w\sigma^2)^{\ell-1} \, \|h_1\|^2, \tag{5}$$

$$\mathbb{E}\big[\|h_\ell - h_1\|^2\big] \;\leq\; \big[(1 + w\sigma^2)^{\ell-1} - 1\big] \, \|h_1\|^2. \tag{6}$$

The right-hand side of (6) matches the linearized identity (4) of Theorem A.2; for the actual ReLU recursion the equality becomes an inequality, with no transparency hypothesis required.

Proof. Let $\mathcal{F}_k := \sigma(\Delta_1, \dots, \Delta_{k-1})$ so that h_k is $\mathcal{F}_k$-measurable and Δ_k is independent of $\mathcal{F}_k$.

Step 1: 1-Lipschitz contraction. Since ReLU is 1-Lipschitz with $\mathrm{ReLU}(0) = 0$, applying coordinatewise gives

$$\|h_{k+1}\|^2 \;\leq\; \|(I_w + \Delta_k)h_k\|^2. \tag{7}$$

Step 2: conditional expectation of the linear step. Expanding $\|h_k + \Delta_k h_k\|^2 = \|h_k\|^2 + 2\langle h_k, \Delta_k h_k\rangle + \|\Delta_k h_k\|^2$ and conditioning on $\mathcal{F}_k$ (so h_k is fixed): $\mathbb{E}[\langle h_k, \Delta_k h_k\rangle \mid \mathcal{F}_k] = \langle h_k, \mathbb{E}[\Delta_k] h_k\rangle = 0$, and $\mathbb{E}[\|\Delta_k h_k\|^2 \mid \mathcal{F}_k] = w\sigma^2\|h_k\|^2$ (sum the per-row Gaussian variance $\sigma^2\|h_k\|^2$ over w rows). Hence

$$\mathbb{E}\big[\|h_k + \Delta_k h_k\|^2 \,\big|\, \mathcal{F}_k\big] \;=\; (1 + w\sigma^2)\,\|h_k\|^2. \tag{8}$$

Step 3: norm bound (5). Combining (7)–(8) gives $\mathbb{E}[\|h_{k+1}\|^2 \mid \mathcal{F}_k] \le (1 + w\sigma^2)\|h_k\|^2$, and the tower rule yields $\mathbb{E}\|h_{k+1}\|^2 \le (1 + w\sigma^2)\mathbb{E}\|h_k\|^2$. Iterating from $\mathbb{E}\|h_1\|^2 = \|h_1\|^2$ proves (5).

Step 4: deviation bound (6). Because $h_1 \ge 0$ coordinatewise, $\mathrm{ReLU}(h_1) = h_1$. The 1-Lipschitz property of ReLU applied coordinatewise yields

$$\begin{aligned}\|h_{k+1} - h_1\|^2 &= \big\|\mathrm{ReLU}\big((I_w + \Delta_k)h_k\big) - \mathrm{ReLU}(h_1)\big\|^2 \\ &\le \|(I_w + \Delta_k)h_k - h_1\|^2 \\ &= \|(h_k - h_1) + \Delta_k h_k\|^2.\end{aligned}$$

Expanding and conditioning on $\mathcal{F}_k$ as in Step 2,

$$\begin{aligned}&\mathbb{E}\big[\|(h_k - h_1) + \Delta_k h_k\|^2 \,\big|\, \mathcal{F}_k\big] \\ &\quad= \|h_k - h_1\|^2 + 2\langle h_k - h_1, \mathbb{E}[\Delta_k] h_k\rangle + w\sigma^2\|h_k\|^2 \\ &\quad= \|h_k - h_1\|^2 + w\sigma^2\,\|h_k\|^2.\end{aligned}$$

Hence $\mathbb{E}\|h_{k+1} - h_1\|^2 \le \mathbb{E}\|h_k - h_1\|^2 + w\sigma^2\,\mathbb{E}\|h_k\|^2$. Setting $a_k := \mathbb{E}\|h_k - h_1\|^2$ and using (5),

$$a_{k+1} \;\le\; a_k + w\sigma^2\,(1 + w\sigma^2)^{k-1}\,\|h_1\|^2.$$

Telescoping with $a_1 = 0$:

$$\begin{aligned}a_\ell \;&\le\; \|h_1\|^2 \cdot w\sigma^2 \sum_{k=1}^{\ell-1} (1 + w\sigma^2)^{k-1} \\ &=\; \|h_1\|^2 \cdot \big[(1 + w\sigma^2)^{\ell-1} - 1\big],\end{aligned}$$

which is (6). □

Remark A.6 (Scope and relation to the existing results). Theorem A.5 requires only $h_1 \ge 0$ coordinatewise, which is automatic in THEORY-INIT (where $h_1 = \mathrm{ReLU}(W_1 x + b_1)$). It removes the $\min_j (h_1)_j \ge c > 0$ strict-positive-margin condition that Theorem A.7 uses to bound $\Pr(\mathcal{T}_\ell^c)$: the L^2-mean preservation no longer hinges on the no-clipping event. The corresponding tightening in the main paper is that the propagation bound on $\mathbb{E}\|h_\ell - h_1\|^2/\|h_1\|^2$ from Theorem A.2 is now an unconditional fact about the actual recursion, not just the linearized one. Theorem A.7 is still needed for the pathwise coupling $h_k = \tilde{h}_k$ used in Theorem A.12; this proposition does not provide that lift.

A.4. Proof of Theorem A.7 (ReLU transparency probability)

Proposition A.7 (Transparency probability). *Suppose* $\min_j (h_1)_j \ge c > 0$ *and* $\|h_1\| \le R$*, and write* $\gamma := 1 + 3\sigma\sqrt{w}$*. Under* (3)*, the failure probability decomposes into a* conditional *bound and an unconditional one.*

(i) Conditional bound. *On the good event* $E := \bigcap_{k=1}^{\ell-1} E_k$ *defined in the proof (coordinate drift* $\le c/2$ *and norm growth* $\le \gamma^{2(k-1)} R^2$ *uniformly in* k*),*

$$\begin{aligned}&\Pr(\textit{transparency fails} \mid E) \\ &\quad\le\; w(\ell - 1)\exp\left(-\frac{c^2}{8\,\sigma^2\,\gamma^{2(\ell-2)} R^2}\right).\end{aligned} \tag{9}$$

(ii) Unconditional bound. *If the deviation-budget precondition*

$$R\,\sigma\,\gamma^{\ell-3}\sqrt{2(\ell - 2)\log\tfrac{4w(\ell-2)}{\eta}} \;\le\; \tfrac{c}{2} \tag{10}$$

holds for some $\eta \in (0, 1)$ *with* $2(\ell - 2)\,e^{-w/2} \le \eta/2$*, then*

$$\begin{aligned}&\Pr(\textit{transparency fails}) \\ &\quad\le\; \eta \;+\; w(\ell - 1)\exp\left(-\frac{c^2}{8\,\sigma^2\,\gamma^{2(\ell-2)} R^2}\right).\end{aligned} \tag{11}$$

Proof. Throughout this proof we write h_k for the linearized iterate $\tilde{h}_k$ defined by (3), since the proposition's hypothesis is stated under that recursion; the no-clipping event $\mathcal{T}_\ell$ is precisely the event on which the actual ReLU recursion coincides pathwise with the linearized one (Theorem A.3), so probabilities computed for $\tilde{h}_k$ transfer directly to the actual ReLU recursion via that coupling. Fix $k \le \ell - 1$ and $j \le w$. Conditional on h_k, $(\Delta_k h_k)_j \sim \mathcal{N}(0, \sigma^2 \|h_k\|^2)$. The probability of clipping at coordinate (k, j) is

$$\Pr\big((h_k)_j + (\Delta_k h_k)_j < 0 \mid h_k\big) = \Phi\Big(-\tfrac{(h_k)_j}{\sigma \|h_k\|}\Big) \le \exp\Big(-\tfrac{(h_k)_j^2}{2\sigma^2 \|h_k\|^2}\Big),$$

using $\Phi(-t) \le e^{-t^2/2}$ for $t > 0$.

Good event E_k. Set $\xi_k := h_k - h_1$. By Prop. A.2, $\mathbb{E}\|\xi_k\|^2 = ((1 + w\sigma^2)^{k-1} - 1)\|h_1\|^2$. Define

$$E_k := \{|(\xi_k)_j| \le c/2 \;\forall j \le w\} \;\cap\; \{\|h_k\|^2 \le \gamma^{2(k-1)} R^2\}.$$

On E_k we have $(h_k)_j \ge c/2$ and $\|h_k\|^2 \le \gamma^{2(k-1)} R^2 \le \gamma^{2(\ell-2)} R^2$ for $k \le \ell - 1$.

Conditional bound. Substituting $(h_k)_j \ge c/2$ and the operator-norm growth bound on $\|h_k\|$ into the Gaussian-tail estimate,

$$\Pr(\text{clipping at } (k, j) \mid E_k) \le \exp\left(-\frac{c^2}{8\,\sigma^2\,\gamma^{2(\ell-2)} R^2}\right).$$

Union bounding over $j \le w$ and $k \le \ell - 1$ gives (9).

Unconditional bound. We bound $\Pr(E^c) \le \eta$ using a martingale tail for the coordinate drift plus a Davidson–Szarek tail for the norm. On the spectral-norm event $\mathcal{E}_{\text{spec}} := \bigcap_{r=1}^{\ell-2}\{\|\Delta_r\|_{\text{op}} \le 3\sigma\sqrt{w}\}$, $\|h_k\| \le R\,\gamma^{k-1}$ and the predictable variance of $(h_k)_j - (h_1)_j = \sum_{q<k}(\Delta_q h_q)_j$ is at most $\sigma^2(k-1)R^2\gamma^{2(k-2)}$. By the sub-Gaussian martingale tail and a union bound over $j \le w$, $k \le \ell - 1$, the deviation event $\{|(\xi_k)_j| \le c/2 \;\forall j, k\}$ holds on $\mathcal{E}_{\text{spec}}$ with failure $\le \eta/2$ provided (10) holds (this matches the threshold in the standard sub-Gaussian tail $2\exp(-t^2/(2V_r)) \le \eta/(2w(\ell-2))$). By Theorem A.11, $\Pr(\mathcal{E}_{\text{spec}}^c) \le 2(\ell-2)e^{-w/2} \le \eta/2$. Together $\Pr(E^c) \le \eta$. Combining with (9) yields (11). ☐

Remark A.8 (Scope: zero coordinates and architectural applicability). Theorem A.7 requires $\min_j (h_1)_j \ge c > 0$ strictly. The default PWL-plus-padding architecture used here produces h_1 with several zero coordinates with positive probability: a shifted-ReLU hinge $\mathrm{ReLU}(c_i^\top x - t_i)$ vanishes whenever $c_i^\top x \le t_i$ (so several theory neurons can be inactive on inputs near the lower end of the breakpoint range), and each padding coordinate $\mathrm{ReLU}(W_P x)_j$ is exactly zero with probability $1/2$ by Gaussian symmetry. The strict-positive-margin condition is needed only for the no-clipping event used to couple the actual ReLU recursion to the linearized one; it is *not* needed for the L^2-mean preservation statement (4), which is unconditional for the actual ReLU recursion as soon as $h_1 \ge 0$ coordinatewise (Theorem A.5).

Remark A.9 (Empirical transparency at initialization). We compute the per-(input, layer, coordinate) transparency rate at $t = 0$ on the held-out test split for the deployed THEORY-INIT networks (4 seeds, $\sigma_\Delta = 10^{-2}$). Table 2 reports the fraction of triples with pre-ReLU activation ≥ 0. On the deployed PWL+padding architecture the rates are 49–80%, with the layer-1 floor at $\sim 50\%$ reflecting the zero-coordinate construction (each padding row is zero with probability $1/2$ by Gaussian symmetry; PWL hinges are zero on inputs below the breakpoint range). For comparison, the same machinery applied to a strictly-positive synthetic $h_1 \sim \text{Uniform}[0.1, 1]^{64}$ gives 100.0% transparency at $\sigma_\Delta = 10^{-3}$ and 99.8% at $\sigma_\Delta = 10^{-2}$, confirming Theorem A.7 in its native regime. The empirical rate quantifies the qualitative scope statement of Theorem A.8: the pathwise lifting in Theorem A.12 (and the corresponding actual-ReLU clause in Cor. 3.3 of the main paper) is therefore approximate (not nearly-sure) on the deployed architecture, while the unconditional L^2-mean preservation in Theorem A.5 continues to apply unmodified.

Table 2. Empirical transparency rate at t=0: fraction of (input, layer, coord) triples with pre-ReLU ≥ 0 for the THEORY-INIT network on the held-out test split. Mean $\pm$ std over 4 seeds. Layers 2–3 are the middle blocks invoked by Theorem A.12 and Cor. 3.3 of the main paper.

Dataset	Layer 1	Layer 2	Layer 3
Yacht	$50.6 \pm 1.2\%$	$70.1 \pm 1.8\%$	$74.0 \pm 1.7\%$
Energy	$51.6 \pm 1.3\%$	$70.0 \pm 3.3\%$	$76.1 \pm 2.4\%$
Concrete	$50.3 \pm 1.8\%$	$74.0 \pm 1.5\%$	$79.7 \pm 1.4\%$
LogP	$49.1 \pm 3.2\%$	$77.5 \pm 2.9\%$	$79.3 \pm 2.6\%$

Remark A.10 (Numerical example). For w=64, σ=10^{-2}, ℓ−1=2, c/R=0.1, the operator-norm factor $\gamma^{2(\ell-2)}$=$(1.24)^2 \approx 1.538$, so the exponent is $c^2/(8\sigma^2\gamma^{2(\ell-2)}R^2) \approx 8.13$ and the worst-case union-bound prefactor $w(\ell-1)e^{-8.13} \approx 0.038$

certifies a no-clipping failure probability of at most $\sim 3.8\%$ in the strict-positive-margin regime. The deployed PWL+padding architecture does not satisfy that precondition (Theorem A.8); the empirical per-coordinate transparency rate (Theorem A.9) is the relevant signal at this σ. For $\sigma{=}10^{-3}$ ($\gamma{=}1.024$, $\gamma^{2(\ell-2)}{\approx}1.049$) the exponent is $c^2/(8\sigma^2\gamma^{2(\ell-2)}R^2) \approx 1192$ and the bound $w(\ell-1)e^{-1192}$ is well below 10^{-200}; the unconditional deviation budget contributes $\eta \ll 10^{-6}$ on top of this.

A.5. Proof of Theorem A.12 (near-isometric middle-block Jacobian)

Lemma A.11 (Spectral norm concentration; Davidson–Szarek)**.** *For $\Delta \in \mathbb{R}^{w\times w}$ with $(\Delta)_{ij} \overset{iid}{\sim} \mathcal{N}(0,\sigma^2)$ and $t>0$,*

$$\Pr\big(\, \|\Delta\|_{\mathrm{op}} \geq \sigma(2\sqrt{w}+t)\big) \leq 2\, e^{-t^2/2}. \tag{12}$$

With $t=\sqrt{w}$: $\Pr(\|\Delta\|_{\mathrm{op}} \geq 3\sigma\sqrt{w}) \leq 2e^{-w/2}$.

Proof. Classical Davidson–Szarek bound (Davidson & Szarek, 2001); see also Vershynin, 2018, Section 7.3. □

Corollary A.12 (Near-isometric Jacobian)**.** *Let $J = \prod_{k=1}^{\ell-1}(I_w + \Delta_k)$ be the linearized middle-layer Jacobian on the transparency event. With probability at least $1-2(\ell-1)e^{-w/2}$,*

$$\sigma_i(J) \in \big[(1-3\sigma\sqrt{w})^{\ell-1},\ (1+3\sigma\sqrt{w})^{\ell-1}\big] \quad \textit{for all } i. \tag{13}$$

Proof. Weyl's perturbation gives $|\sigma_i(I+\Delta_k)-1| \leq \|\Delta_k\|_{\mathrm{op}}$. On $\|\Delta_k\|_{\mathrm{op}} \leq 3\sigma\sqrt{w}$, $\sigma_i(I+\Delta_k) \in [1-3\sigma\sqrt{w}, 1+3\sigma\sqrt{w}]$. Submultiplicativity for invertible factors yields the product form. A union bound over $\ell-1$ layer events gives the stated probability. □

Remark A.13 (Numerical values for the small-MLP architecture). For $w=64$, $\ell-1=2$:

- $\sigma = 10^{-3}$: $3\sigma\sqrt{w} = 0.024$, per-layer $\sigma_i \in [0.976, 1.024]$, product $[0.953, 1.049]$, probability $\geq 1-4e^{-32} \approx 1-5.07\times 10^{-14}$.
- $\sigma = 10^{-2}$: $3\sigma\sqrt{w} = 0.24$, per-layer $\sigma_i \in [0.76, 1.24]$, product $[0.578, 1.538]$.

Remark A.14 (Op-norm distance to identity). The bound on singular values controls $\|J\|_{\mathrm{op}}$ and $\|J^{-1}\|_{\mathrm{op}}$ but does not by itself constrain how far J is from I_w in operator norm; a near-isometric J may still scramble coordinates. On the same event of Theorem A.12, $\|J - I_w\|_{\mathrm{op}} \leq \prod_{k=1}^{\ell-1}(1+\|\Delta_k\|_{\mathrm{op}}) - 1 \leq (1+3\sigma\sqrt{w})^{\ell-1}-1$, which at $w=64$, $\sigma=10^{-2}$, $\ell-1=2$ is $(1.24)^2-1 \approx 0.538$, and at $\sigma = 10^{-3}$ is ≈ 0.0486. The latter is tight enough to prevent coordinate scrambling at $\sigma=10^{-3}$; at $\sigma=10^{-2}$ the bound is informative but loose, and the index-preservation claim relies more directly on the explicit $I+\Delta$ structure of each factor than on the singular-spectrum bound alone.

A.6. First-order regime and safe-depth bound

Corollary A.15 (First-order signal preservation)**.** *For $w\sigma^2 \ll 1$,*

$$\mathbb{E}[\|h_\ell - h_1\|^2] = (\ell-1)\, w\sigma^2\, \|h_1\|^2 + O\big((\ell-1)^2(w\sigma^2)^2\, \|h_1\|^2\big). \tag{14}$$

The linear term is accurate within 1% whenever $(\ell-1)\, w\sigma^2 \leq 0.02$.

Proof. Binomial expansion of $(1+w\sigma^2)^{\ell-1}-1$; remainder $\leq \binom{\ell-1}{2}(w\sigma^2)^2(1+w\sigma^2)^{\ell-3}$. □

Corollary A.16 (Safe-depth bound)**.** *For $\varepsilon,\delta \in (0,1)$, an auxiliary failure budget $\eta \in (0,\delta)$, $c \leq \min_j (h_1)_j$, $R \geq \|h_1\|$, and $\gamma := 1+3\sigma\sqrt{w}$, requiring $\mathbb{E}\|h_\ell - h_1\|^2 \leq \varepsilon^2\|h_1\|^2$ and $\Pr(\textit{transparency}) \geq 1-\delta$ simultaneously, it suffices that all three of the following hold:*

$$(1+w\sigma^2)^{\ell-1} \ \leq\ 1+\varepsilon^2, \tag{15}$$

$$R\sigma\gamma^{\ell-3}\sqrt{2(\ell{-}2)\log\tfrac{4w(\ell-2)}{\eta}} \ \leq\ \tfrac{c}{2}, \tag{16}$$

$$\eta + w(\ell{-}1)\exp\Big(-\tfrac{c^2}{8\sigma^2\gamma^{2(\ell-2)}R^2}\Big) \ \leq\ \delta, \tag{17}$$

corresponding to (i) norm growth (Theorem A.2), (ii) the deviation budget (10), and (iii) the transparency union bound (11). (15) solves explicitly to $\ell-1 \leq \ln(1+\varepsilon^2)/\ln(1+w\sigma^2) \approx \varepsilon^2/(w\sigma^2)$; (16) and (17) are implicit in σ, ℓ and typically (16) (the deviation budget) binds first because it grows as $\gamma^{\ell-3}\sqrt{\ell-2}$ while (17) only enters through a logarithm. At the small-MLP default ($w=64$, $\sigma=10^{-2}$, $c/R=0.1$, $\ell-1=2$) all three are non-binding numerically and reduce to the bound quoted in Theorem A.10; the deviation budget is the structurally tightest constraint as ℓ grows.

A.7. Connection to dynamical isometry

The dynamical-isometry line (Saxe et al., 2014; Pennington et al., 2017) shows that for i.i.d. Gaussian initialization $W = \sigma G$, the input–output Jacobian's singular spectrum diverges in the deep limit. Strict dynamical isometry is achievable only via measure-zero subgroups (e.g. orthogonal initialization). Theorem A.12 avoids this: the Jacobian is a small Gaussian perturbation of an exact isometry (I_w), with perturbation magnitude controlled directly by σ. This is middle-block near-isometry by structural choice rather than by limit-tuning.

A.8. Connection to the permutation orbit

This subsection bridges Theorems A.2, A.7 and A.12 and the empirical pre-alignment finding.

THE PERMUTATION ORBIT AT INITIALIZATION

Definition A.17 (Permutation orbit at layer ℓ). For an MLP with hidden width w, a *neuron permutation* $\pi = (P_1, \ldots, P_{L-1})$ is a tuple of $w \times w$ permutation matrices. The action on parameters is $W_1 \mapsto P_1 W_1$, $W_\ell \mapsto P_\ell W_\ell P_{\ell-1}^\top$ for $\ell = 2, \ldots, L-1$, $W_L \mapsto W_L P_{L-1}^\top$, and $b_\ell \mapsto P_\ell b_\ell$. The set $\mathcal{O}(\theta) = \{\pi \cdot \theta\}$ for $\pi \in S_w^{L-1}$ is the *permutation orbit* of θ.

Remark A.18 (How THEORY-INIT narrows the orbit). At initialization, the layer-1 weights of THEORY-INIT are N *prescribed* physics rows in fixed positions $\{1, \ldots, N\}$ plus $w - N$ small-random padding rows in $\{N+1, \ldots, w\}$. A layer-1 permutation P_1 that swaps two physics rows, applied without the corresponding inverse column permutation in the next layer, changes the function the network represents. Hence the layer-1 stabilizer at initialization is S_{w-N} (permuting only the padding), and the corresponding layer-1 orbit of placements has size $w!/(w-N)!$. For middle layers, the identity-plus-noise structure makes W_ℓ a near-identity matrix; the layer-ℓ stabilizer at initialization is again S_{w-N}.

COST-MATRIX TRACE-DOMINANCE FOR THEORY–THEORY PAIRS

Git Re-Basin (Ainsworth et al., 2023) aligns (θ_A, θ_B) layer-by-layer by solving the linear-assignment problem $P_\ell^* = \arg\max_{P \in S_w} \operatorname{tr}(P\, M_\ell)$. Following Ainsworth et al. (2023, Algorithm 1), the layer-ℓ matching cost is

$$M_\ell \;=\; W_\ell^A (W_\ell^B)^\top \;+\; (W_{\ell+1}^A)^\top W_{\ell+1}^B \;+\; b_\ell^A (b_\ell^B)^\top, \tag{18}$$

with the maximizer computed by a standard linear-sum-assignment (Hungarian) solver.

Lemma A.19 (Trace-dominance of the layer-1 cost matrix at init)**.** *Let θ_A, θ_B be two* THEORY-INIT *initializations with the same theory $\hat{f}$ but different padding seeds. Write $W_1^A = T + R_A$, $W_1^B = T + R_B$, $b_1^A = \beta + \rho_A$, $b_1^B = \beta + \rho_B$, where (T, β) is the deterministic theory portion (rows $1{:}N$ of T nonzero, rows $N{+}1{:}w$ zero; $\beta_i = 0$ for $i > N$) and R_A, R_B, ρ_A, ρ_B are the small-random padding (top N rows of $R_\bullet$ zero; bottom rows iid $\mathcal{N}(0, \alpha_1^2 \cdot 2/d_{\text{in}})$ with $\alpha_1 \le 0.1$; $\rho_i = 0$ for $i \le N$). For middle layers, $W_\ell^A = I + \Delta_\ell^A$ and $W_\ell^B = I + \Delta_\ell^B$ with $\Delta_\ell^\bullet$ iid $\mathcal{N}(0, \sigma^2)$, $\sigma \le 10^{-2}$. Let M_1 be the cost matrix* (18) *at layer* 1*, evaluated at initialization with the layer-*2 *permutation set to identity. Then:*

(a) (Deterministic block.) *The top-$N \times N$ submatrix of $\mathbb{E}[M_1]$ equals $G_{\det} := TT^\top + \beta\beta^\top + I_N$, where $G := TT^\top + \beta\beta^\top$ is symmetric positive semi-definite (PSD).*

(b) (Trace dominance.) *For every permutation $\pi \in S_N$, $\sum_{i=1}^N (G_{\det})_{i,\pi(i)} \le \operatorname{tr}(G_{\det})$, with strict inequality whenever $\pi \neq \mathrm{id}$ does not stabilize the eigenspaces of $G_{\det}$. Define the assignment gap $g_* := \operatorname{tr}(G_{\det}) - \max_{\pi \neq \mathrm{id}} \sum_{i=1}^N (G_{\det})_{i,\pi(i)}$, so identity is strictly optimal iff $g_* > 0$ (the Yacht example below realizes a gap of* 3.25 *on G alone, widening to* 5.25 *on $G_{\det}$; the LogP swap-vs-identity gap on G is $4M^2 = 8$).*

(c) (Padding block.) *The bottom-$(w{-}N) \times (w{-}N)$ block of $\mathbb{E}[M_1]$ has off-diagonal entries with mean* 0 *and diagonal mean* 1 *(the layer-*2 *identity contribution; the cross-padding term $\mathbb{E}\langle R_{i,:}^A, R_{i,:}^B\rangle = 0$ by independence and zero mean); with probability $\ge 1 - O(e^{-w})$ the diagonal dominates the row-sum (Bernstein).*

Proof. (a) Substituting $W_1^A = T + R_A$, $W_1^B = T + R_B$ into (18) and using $\mathbb{E}[R_\bullet] = \mathbb{E}[\rho_\bullet] = 0$ together with the block structure (top N rows of $R_\bullet$ are zero; bottom $w{-}N$ rows of T are zero) gives, on indices $i, j \le N$,

$$\mathbb{E}[M_1]_{ij} \;=\; \langle T_{i,:}, T_{j,:}\rangle \;+\; \mathbb{E}\big[(W_2^A)^\top W_2^B\big]_{ij} \;+\; \beta_i \beta_j.$$

For $W_2^\bullet = I + \Delta_2^\bullet$ with Δ_2^A, Δ_2^B independent and mean-zero, $\mathbb{E}[(I+\Delta_2^A)^\top (I+\Delta_2^B)] = I + 0 + 0 + \mathbb{E}[(\Delta_2^A)^\top]\mathbb{E}[\Delta_2^B] = I_w$, so the middle term contributes δ_{ij} on the top block. Hence the top block of $\mathbb{E}[M_1]$ is $G + I_N$ with $G = TT^\top + \beta\beta^\top$. Both $TT^\top$ (a Gram matrix) and $\beta\beta^\top$ (rank-1 outer of a real vector with itself) are PSD, so G is PSD. The layer-2 noise contributes only at second moments, not in expectation, so $\mathbb{E}[M_1]$ on the top block is exactly $G + I_N$.

(b) Write $H = VV^\top$ with $V \in \mathbb{R}^{N \times N}$ (any PSD H admits such a Gram factorization, with V taken to be a square root of H); let v_i denote the i-th row of V. Then $H_{ij} = \langle v_i, v_j\rangle$, so $\sum_i H_{i,\pi(i)} = \sum_i \langle v_i, v_{\pi(i)}\rangle$. Cauchy–Schwarz coordinate-wise

gives $\langle v_i, v_{\pi(i)}\rangle \le \frac{1}{2}(\|v_i\|^2 + \|v_{\pi(i)}\|^2)$, and summing over i (using that π is a bijection so $\sum_i \|v_{\pi(i)}\|^2 = \sum_i \|v_i\|^2$) yields

$$\sum_{i=1}^{N} H_{i,\pi(i)} \;\le\; \sum_{i=1}^{N} \|v_i\|^2 \;=\; \sum_{i=1}^{N} H_{ii} \;=\; \mathrm{tr}(H),$$

with strict inequality whenever $\pi \ne \mathrm{id}$ moves an index i to a $\pi(i)$ with $v_{\pi(i)} \ne \pm v_i$ (e.g. for H with distinct diagonal entries that occur as $\|v_i\|^2$, identity is the unique optimum). (Equivalently in singular-value form: every permutation matrix P_π has all singular values equal to 1, so von Neumann's trace inequality gives $\mathrm{tr}(P_\pi H) \le \sum_i \sigma_i(P_\pi)\sigma_i(H) = \mathrm{tr}(H)$ for PSD H.) Applied to $H = G_{\mathrm{det}}$, this proves identity is optimal among permutations.

(c) The bottom block of M_1 has entries equal to sums of products of independent mean-zero Gaussians plus a deterministic diagonal contribution from layer 2 (δ_{ij}). Bernstein's inequality, applied row-wise (length $w - N$), bounds the off-diagonal row-sum by $O(\alpha_1^2\sqrt{(w-N)/d_{\mathrm{in}}})$ and the layer-2 off-diagonal contribution by $O(\sigma\sqrt{w \log w} + w\sigma^2)$, with probability $\ge 1 - O(e^{-w})$. Provided

$$\alpha_1^2\,\sqrt{(w-N)/d_{\mathrm{in}}} \;+\; \sigma\sqrt{w\log w} \;+\; w\sigma^2 \;\ll\; 1,$$

this is $\ll 1 \le M_{ii}$, so the diagonal dominates the row-sum. At the small-MLP default ($w = 64$, $N \le 4$, $\alpha_1 = 0.1$, $d_{\mathrm{in}} = 6$, $\sigma = 10^{-2}$) the LHS is $\approx 0.032 + 0.16 + 0.0064 \approx 0.20$, comfortably below 1. Cf. (Burkard et al., 2009) for general sensitivity results on assignment problems. □

Remark A.20 (Numerical verification on Yacht). Substituting the Yacht physics rows (Section C.2) into $G = TT^\top + \beta\beta^\top$ yields, with $\mathrm{tr}(G) = 61.5$,

$$G \;=\; \begin{pmatrix} 25.25 & -14.25 & 10 & 19.5 \\ -14.25 & 11.25 & -6 & -13.5 \\ 10 & -6 & 8 & 8 \\ 19.5 & -13.5 & 8 & 17 \end{pmatrix}.$$

Strict pointwise diagonal dominance *fails* ($|G_{0,3}| = 19.5 > G_{3,3} = 17$); nevertheless an exhaustive sweep over all $4! = 24$ permutations confirms $\mathrm{tr}(G) = 61.5$ strictly exceeds every other $\sum_i G_{i,\pi(i)}$ (next-best $\pi = (3,1,2,0)$: 58.25, gap 3.25 on G; the deterministic block $G_{\mathrm{det}} = G + I_N$ adds $N = 4$ to the identity sum and $\sum_i \delta_{i,\pi(i)} = 2$ to the next-best, widening the gap to 5.25). For Energy and Concrete (Sections C.3 and C.4) the deterministic block G is diagonal, so identity-optimality is immediate. For LogP (Section C.5, rows 0 and 2 are exact negatives), pointwise diagonal dominance is at the boundary, but the *signed* cost still strictly favors the identity: the swap that exchanges rows 0 and 2 replaces $G_{00} + G_{22} = 2M^2 + 0.5$ in the trace with $G_{20} + G_{02} = -2M^2 + 0.5$, a gap of $4M^2 = 8$ (the bias-variance terms cancel). This is precisely the gap that pointwise diagonal dominance misses but the PSD/trace bound captures.

Corollary A.21 (Identity is Hungarian-optimal at layer 1)**.** *Under Theorem A.19 with $g_* > 0$, with probability $\ge 1 - O(e^{-w})$ the linear-assignment maximizer of $\mathrm{tr}(P\,M_1)$ over $P \in S_w$ is the identity (up to a free permutation of the $w - N$ padding indices, which leaves the function invariant by Theorem A.18). It suffices that the random perturbation of the assignment cost is $< g_*/2$, which holds with probability $\ge 1 - O(e^{-w})$ for w moderately large.*

Proof. Decompose $\mathrm{tr}(P\,M_1)$ into top-N and bottom-$(w - N)$ contributions plus cross terms. The cross terms vanish in expectation (top rows of $R_\bullet$ are zero; bottom rows of T are zero). The top contribution is uniquely maximized by identity by Theorem A.19(b); concentration of the random part is $O(\alpha_1\sqrt{2/d_{\mathrm{in}}})$ per entry, hence smaller than the trace gap with probability $\ge 1 - O(e^{-w})$. The bottom block is handled by Theorem A.19(c). □

Remark A.22 (Identity-permutation observation). Theorem A.21 predicts that for theory–theory pairs the Hungarian-optimal layer-1 permutation is identity (on the physics rows) with high probability. We do not formally prove an analogous statement for middle or trained-time layers (the layer-1 trace gap of g_*=3.25 on Yacht comes from the N theory rows; middle layers carry only the smaller $I_w + O(\sigma^2)$ block from $W_\ell = I + \Delta_\ell$, and the layer-2 coupling $\mathbb{E}[(I + \Delta_2^A)^\top (I + \Delta_2^B)] = I_w$ only reinforces identity-optimality at layer 1). Empirically, Git Re-Basin returns the identity permutation on THEORY-INIT–THEORY-INIT pairs at every layer and after training (median aligned-to-naive distance ratio $d_{\mathrm{al}}/d_{\mathrm{nv}}$=0.996, n=24 pairs across 4 datasets), which is an empirical observation, not a theorem. The aligned-to-naive *barrier* ratio pooled across all 24 pairs is also ≈1.000, but this pooled median masks modest per-dataset barrier drops (medians: Yacht 0.71, Concrete 0.79, LogP 0.86, Energy 0.92): because a barrier is a maximum of the loss over the interpolation path, aligning to a near-identical frame can still lower it slightly even when the distance is essentially unchanged. The robust, scale-free statement is that the distance is unchanged (identity permutation); we do not claim the barrier is unchanged to within a percent on every dataset.

Remark A.23 (Frobenius-distance form). If $\|W_1^A - W_1^B\|_F \le \varepsilon\,\|W_1^A\|_F$ and the smallest singular value of the top-N block of W_1^A is bounded below by $\tau_{\min}$, then for ε small the Hungarian-optimal permutation is the identity. For THEORY-INIT,

ε measures only the padding noise (deterministic theory rows are seed-invariant), giving a hard upper bound on ε that is independent of the data.

A.9. Summary of bounds for the small-MLP architecture

Table 3 consolidates the numerical values that appear throughout §A.2–§A.5 for the small-MLP default ($w = 64$, $L = 4$, $\ell - 1 = 2$, $\sigma \in \{0, 10^{-3}, 10^{-2}\}$).

Table 3. Summary of bounds for the small-MLP architecture ($w = 64$, $L = 4$, $\ell - 1 = 2$).

Quantity	$\sigma = 0$	$\sigma = 10^{-3}$	$\sigma = 10^{-2}$
Relative signal deviation $\sqrt{\mathbb{E}\|h_3 - h_1\|^2/\|h_1\|^2}$	0%	1.13%	11.33%
Per-layer Jacobian sing. values	$\{1\}$	$[0.976, 1.024]$	$[0.76, 1.24]$
Product Jacobian sing. values	$\{1\}$	$[0.953, 1.049]$	$[0.578, 1.538]$
$\Pr(\text{transparency holds} \mid E)$, $c/R \geq 0.1$	1	$\geq 1 - 10^{-200}$	≥ 0.96
$\Pr(\text{transparency holds})$ unconditional, $c/R \geq 0.1$	1	$\geq 1 - 10^{-200}$	≥ 0.96
$\Pr(\text{near-isometry}, w/2 \text{ tail})$	1	$\geq 1 - 4e^{-32}$	$\geq 1 - 4e^{-32}$

B. Method details

B.1. Stage 1: theory extraction

THEORY-INIT computes a theory approximation $\hat{f}$ of the target function on the input domain. We use one of three sources, in increasing order of domain specificity:

- **Taylor expansion** (synthetic 1D analytic targets): $\hat{f}(x) = \sum_{k=0}^{K} a_k x^k / k!$ at the center of the input domain.
- **Chebyshev nodes** (harder targets, including high-curvature 1D functions and 2D additive/product decompositions): $\hat{f}$ is a Chebyshev interpolant of degree K on the input domain.
- **Domain models** (real-data tasks): Crippen atomic contributions for molecular LogP (Wildman & Crippen, 1999), Froude scaling for hull resistance, Abrams' law for concrete strength, BPM-velocity scaling for airfoil noise.

B.2. Stage 2: PWL physics-neuron construction

We discretize $\hat{f}$ at N breakpoints $\{t_i\}$ with values $v_i = \hat{f}(t_i)$ and slope changes $s_i = (v_{i+1} - v_i)/(t_{i+1} - t_i) - (v_i - v_{i-1})/(t_i - t_{i-1})$. The first N neurons of layer 1 are prescribed to compute $\mathrm{ReLU}(x \cdot u_i - t_i)$ where u_i is an input direction (typically a fixed feature index for 1D targets, or a feature combination for domain models). Concretely, with input $x \in \mathbb{R}^{d_{\text{in}}}$, the first N *theory rows* are deterministic and the remaining $w - N$ *padding rows* are small-random:

$$W_{1,i,:} = u_i^\top, \quad b_{1,i} = -t_i, \quad i = 1, \dots, N, \tag{19}$$

$$W_{1,j,:} \sim \mathcal{N}\big(0,\, \alpha_1^2 \cdot 2/d_{\text{in}}\big), \quad b_{1,j} = 0, \quad j = N{+}1, \dots, w, \tag{20}$$

with $\alpha_1 \leq 0.1$ for real-data tasks.

Output-layer alignment for PWL function. For the function-zoo experiments where the target is $\hat{f}$ exactly, the slope-change values $\{s_i\}$ are written into the appropriate output-layer column (alongside small-random padding). For UCI and molecular LogP the output layer is fully random ($\mathcal{N}(0, \alpha_{\text{out}}^2 \cdot 2/w)$, $\alpha_{\text{out}} = 0.1$); the network learns the linear combination of theory features during training.

B.3. Stage 3: identity-plus-noise middle layers

Middle layers $\ell = 2, \dots, L - 1$ are

$$W_\ell = I_w + \Delta_\ell, \qquad b_\ell = 0, \qquad (\Delta_\ell)_{ij} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\Delta^2),$$

with the entries of Δ_ℓ drawn independently across ℓ. We use three regimes for σ_Δ:

- **Dead** ($\sigma_\Delta = 0$): exact identity. Used when the theory $\hat{f}$ is already exact on the target domain (e.g., synthetic 1D Taylor / Chebyshev targets).
- **Alive** ($\sigma_\Delta = 10^{-3}$): used for compositional theory constructions where the breakpoints are dense.
- **Alive-high** ($\sigma_\Delta = 10^{-2}$): used for real-data experiments (UCI, molecular LogP).

B.4. Hyperparameters

Table 4 lists the per-dataset THEORY-INIT hyperparameters used throughout the empirical sections.

Table 4. Per-dataset THEORY-INIT hyperparameters used for the LMC experiments. N_{phys}: number of physics neurons. α_1: layer-1 padding scale. σ_Δ: middle-layer noise. α_{out}: output-layer scale. The novelty/9-method LogP runs use a different padding scale, ρ=0.3 (Section C).

Dataset	N_{phys}	α_1	σ_Δ	α_{out}	Theory source
Yacht (UCI)	4	0.1	10^{-2}	0.1	Froude + prismatic
Energy (UCI)	3	0.1	10^{-2}	0.1	envelope + glazing + compactness
Concrete (UCI)	2	0.1	10^{-2}	0.1	Abrams' law + log-age
Airfoil (UCI)	2	0.1	10^{-2}	0.1	BPM velocity^5 + Strouhal
Kin8nm (UCI)	3	0.1	10^{-2}	0.1	cumul. kinematic + proximal joint
Molecular LogP	3	0.1	10^{-2}	0.1	Crippen atomic contributions (Wildman & Crippen, 1999)
Synthetic (1D Taylor)	varies	—	0	set by s_i	Taylor coefficients
Synthetic (1D Cheb.)	varies	—	10^{-3}	set by s_i	Chebyshev nodes

B.5. Architecture

The LMC experiment uses architecture $[d_{\text{in}}, 64, 64, 64, 1]$ (three hidden layers of width 64) uniformly across all four small-MLP datasets (Yacht, Energy, Concrete, LogP). The small CNN scale-up uses the architecture in §D.7.

C. Theory-init weight derivation per dataset

This appendix documents, dataset by dataset, the explicit construction of the layer-1 weight matrix $W_1 \in \mathbb{R}^{w \times d_{\text{in}}}$, the bias vector $b_1 \in \mathbb{R}^w$, and the padding scheme used for the remaining $w - N_{\text{phys}}$ rows of layer 1. The middle layers $\ell = 2, \ldots, L-1$ and the output layer are constructed identically across all datasets and are stated once in §C.1. All numerical constants below are stated as implemented.

C.1. Shared middle and output layers

For all tabular datasets (UCI yacht / energy / concrete, MoleculeNet Lipophilicity) the construction of layers $\ell = 2, \ldots, L-1$ uses the *alive-high* mode:

$$W_\ell = \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} + \Delta_\ell, \quad b_\ell = 0,$$

$$(\Delta_\ell)_{ij} \sim \mathcal{N}(0, \sigma_\Delta^2), \quad \sigma_\Delta = 10^{-2}, \quad (21)$$

where $k = \min(n_\ell, n_{\ell+1})$ and the I_k block is the upper-left identity (no padding when $n_\ell = n_{\ell+1}$, which is the default). The output layer is small Gaussian:

$$W_{\text{out},ij} \overset{\text{iid}}{\sim} \mathcal{N}\Big(0, \tfrac{2}{n_{L-1}} \cdot 0.1^2\Big), \qquad b_{\text{out}} = 0. \quad (22)$$

(22) gives a standard deviation of $0.1\sqrt{2/w}$, i.e. $\alpha_{\text{out}} = 0.1$ in the notation of Table 4.

C.2. UCI Yacht Hydrodynamics

Domain knowledge (naval architecture). Residuary resistance is dominated by wave-making resistance, which scales roughly as Fr^4 below hull speed ($\text{Fr} < 0.4$) and rises sharply above; the prismatic coefficient C_p modulates wave-making resistance. Features are normalized to $[-1, 1]$: x_0 longitudinal-position, x_1 prismatic, x_2 length/displacement, x_3 beam/draught, x_4 length/beam, x_5 Froude. Number of physics neurons $N_{\text{phys}} = 4$, $d_{\text{in}} = 6$.
The layer-1 weights are zero except for the four physics entries:

$$\begin{aligned}
&W_{1,0,5} = 5.0, && b_{1,0} = 0.5 && \text{(main Froude)},\\
&W_{1,1,5} = -3.0, && b_{1,1} = 1.5 && \text{(low-Froude)},\\
&W_{1,2,1} = W_{1,2,5} = 2.0, && b_{1,2} = 0.0 && (C_p \times \text{Fr}),\\
&W_{1,3,5} = 4.0, && b_{1,3} = -1.0 && \text{(high-Fr spread)}.
\end{aligned}$$

Padding rows $j = 4, \ldots, w-1$ are filled with i.i.d. Gaussian

$$W_{1,j,i} \sim \mathcal{N}\Big(0, \tfrac{2}{6} \cdot 0.1^2\Big), \qquad b_{1,j} = 0,$$

i.e. $\alpha_1 = 0.1$ in the notation of Table 4.

C.3. UCI Energy Efficiency

Domain knowledge (ISO 13790 envelope physics). Heating load is dominated by conductive losses through walls, roof and glazing. Glazing has $\sim$10× higher U-value than wall; higher relative compactness lowers loss; orientation has near-zero effect. Features (normalized): x_0 relative compactness, x_1 surface area, x_2 wall area, x_3 roof area, x_4 overall height, x_5

$$K_0 = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}, K_1 = \begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix}, K_2 = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{pmatrix}, K_3 = G - \bar{G}, K_4 = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{pmatrix}, K_5 = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{pmatrix}, K_6 = \begin{pmatrix} 0 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & 1 & 0 \end{pmatrix}, K_7 = \begin{pmatrix} -1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

orientation, x_6 glazing area, x_7 glazing-area distribution. Number of physics neurons $N_{\mathrm{phys}} = 3$, $d_{\mathrm{in}} = 8$.
The layer-1 weights are zero except for the three physics entries:

$$\begin{aligned}
&W_{1,0,2} = 2.0,\ W_{1,0,3} = 1.5, && b_{1,0} = 1.0 && \text{(envelope loss)},\\
&W_{1,1,6} = 5.0,\ W_{1,1,1} = 1.0, && b_{1,1} = 0.5 && \text{(glazing dom.)},\\
&W_{1,2,0} = -3.0,\ W_{1,2,4} = 1.5, && b_{1,2} = 1.0 && \text{(compactness)}.
\end{aligned}$$

The orientation feature (x_5) is intentionally given zero weight, encoding the prior that orientation contributes little to heating load. Padding rows $j = 3, \ldots, w-1$ use $W_{1,j,i} \sim \mathcal{N}(0, (2/8) \cdot 0.1^2)$ and $b_{1,j} = 0$.

C.4. UCI Concrete Compressive Strength

Domain knowledge (Abrams' law). Compressive strength is governed by the cement/water mass ratio, with log-of-age as a slow strength-gain modifier. Features (normalized): x_0 cement, x_1 slag, x_2 fly-ash, x_3 water, x_4 superplasticizer, x_5 coarse aggregate, x_6 fine aggregate, x_7 age. Number of physics neurons $N_{\mathrm{phys}} = 2$, $d_{\mathrm{in}} = 8$.
The layer-1 weights are zero except for the two physics entries:

$$\begin{aligned}
&W_{1,0,0} = 2.0,\ W_{1,0,3} = -2.0, && b_{1,0} = 1.0 && \text{(Abrams ratio)},\\
&W_{1,1,7} = 3.0, && b_{1,1} = 1.0 && \text{(log-age)}.
\end{aligned}$$

Padding rows $j = 2, \ldots, w-1$ use $W_{1,j,i} \sim \mathcal{N}(0, (2/8) \cdot 0.1^2)$ and $b_{1,j} = 0$.

C.5. MoleculeNet Lipophilicity (Crippen atomic contributions)

Domain knowledge. The Wildman–Crippen LogP model (Wildman & Crippen, 1999) decomposes molecular LogP as a sum of atom-type contributions $\mathrm{LogP} \approx \sum_i n_i a_i$, with 68 atom types in the original publication. The pre-processing pipeline discovers the unique contribution values appearing in the training molecules (after H atoms are added by RDKit) and filters out atom types that appear in fewer than 10 molecules; this gives $d_{\mathrm{in}} \lesssim 68$ unique types. The novelty-ablation and 9-method runs apply a $\mathrm{min\_count}$=10 atom-type filter, giving d_{in}=55 and architecture $[55, 128, 64, 1]$. The LMC summary (Table 1) uses the unfiltered d=64 vocabulary instead. Number of physics neurons $N_{\mathrm{phys}} = 3$ for the main *theory* family (a, b, d, e, f); $N_{\mathrm{phys}} = 2$ for the c control variant, which uses random directions.
Let $\mathbf{c} \in \mathbb{R}^{d_{\mathrm{in}}}$ denote the vector of Crippen contributions ordered by feature index, and define the row-magnitude constant

$$M = \sqrt{2/d_{\mathrm{in}}} \cdot \sqrt{d_{\mathrm{in}}} \cdot \beta = \sqrt{2} \cdot \beta, \tag{23}$$

where β is the variant-dependent physics-row magnitude (β=1 for the main and control variants a, b, c; β=2 for d, f; β=3 for e). The three physics rows are: (i) the Crippen direction, $W_{1,0,:} = M\,\mathbf{c}/\|\mathbf{c}\|_2$ with $b_{1,0} = 0.5$; (ii) its magnitude, $W_{1,1,:} = M\,|\mathbf{c}|/\|\mathbf{c}\|_2$ with $b_{1,1} = 0.5$; (iii) its negation $W_{1,2,:} = -M\,\mathbf{c}/\|\mathbf{c}\|_2$ with $b_{1,2} = 0.5$ (used together with (i) for ReLU-based $|\cdot|$). Padding rows $j = 3, \ldots, w-1$ are filled with $W_{1,j,i} \sim \mathcal{N}(0, (2/d_{\mathrm{in}}) \cdot \rho^2)$ and $b_{1,j} = 0$, where ρ is variant-specific: ρ=1.0 for variants a, ρ=0.3 for b / c / e, ρ=0.5 for d / f. The main-text b variant therefore uses $(\beta, \rho) = (1.0, 0.3)$, exactly as quoted in the main paper.

C.6. Convolutional theory-init: small CNN on MNIST and CIFAR-10

The convolutional case prescribes only the first conv layer; the per-channel CIFAR variant adjusts the channel dimension only.
The eight 3×3 physics filters. The physics-filter stack is an $(8, 3, 3)$ tensor: Sobel-X, Sobel-Y, Laplacian, zero-mean Gaussian, two DoG, and two diagonal kernels, where $G = (1, 2, 1; 2, 4, 2; 1, 2, 1)/16$ is the standard 3×3 binomial Gaussian kernel and $\bar{G}$ its mean. Each kernel is rescaled to Frobenius norm $\sqrt{2/9}$ (the He standard for an in-fan of $9 = 3 \times 3$):

$$\widetilde{K}_k = K_k \cdot \frac{\sqrt{2/9}}{\|K_k\|_F}, \qquad k = 0, \ldots, 7. \tag{24}$$

MNIST and CIFAR (averaged variant). Conv1 has shape $(C_{\mathrm{out}}, c_{\mathrm{in}}, 3, 3)$ with $C_{\mathrm{out}} = 16$ (MNIST) or 32 (CIFAR-10) and $c_{\mathrm{in}} = 1$ (MNIST) or 3 (CIFAR-10). The first $N_{\mathrm{phys}} = 8$ output channels share the physics filter *averaged across input channels*:

$$W_{\mathrm{conv1},k,c,:,:} = \frac{\widetilde{K}_k}{c_{\mathrm{in}}}, \quad k = 0, \ldots, 7,\ c = 0, \ldots, c_{\mathrm{in}} - 1. \tag{25}$$

Padding rows $k = 8, \ldots, C_{\mathrm{out}} - 1$ are i.i.d. Gaussian $\mathcal{N}(0, 2/(c_{\mathrm{in}} \cdot 9))$ (standard He). Conv1 biases are zero.
CIFAR-10 per-channel variant. In the per-channel variant, conv1 has 32 output channels and 24 of them are physics-

Table 5. Explicit Stage-1 prescription sources from constructive approximation (Yarotsky, 2017; Elbrächter et al., 2021). Width and depth are for the stated construction; m is the refinement count, n the polynomial degree, ρ_E the Bernstein-ellipse parameter of the analytic continuation. The first two blocks are verified numerically against their declared bounds.

target	construction	width	depth	sup error
$\lvert x\rvert$, $\max(x,y)$, clamp, hat, payoffs	exact ReLU identities	≤ 3	2	0
sawtooth (2^s pieces)	s-fold hat composition	3	$s{+}1$	0
x^2 on $[0,1]$	hat-telescope accumulator	3	$m{+}1$	$4^{-(m+1)}$
xy on $[-D,D]^2$	polarization, two squaring tracks*	6	$m{+}2$	$D^2\,2^{-2m-1}$
$\sum_{k\le n} a_k x^k$	Horner chain of multiplications	≤ 9	$O(nm)$	coefficient-controlled
$f\in C[a,b]$, analytic	Chebyshev $\to$ monomial $\to$ Horner	≤ 9	$O(nm)$	$\frac{2M}{\rho_E^n(\rho_E-1)}+\varepsilon_{\text{net}}$
compositions $h\circ g$ (e.g. e^{-x^2}, Φ, BS price)	chain the blocks	≤ 12	additive	$L_h\,\varepsilon_g+\varepsilon_h$

*The two squaring tracks accumulate u^2 and v^2 whose difference is signed; an implementation must keep that difference out of the ReLU path. The block-diagonal width-6 form (one accumulator per track, subtraction at the linear output) is verified correct for all sign combinations; a single shared clipped accumulator fails on the $xy<0$ quadrant.

prescribed: filter k acting on *exactly one* input channel,

$$W_{\text{conv1},\,8c_{\text{in}}+k,\,c,\,:,:} = \widetilde{K}_k\,\delta_{c=c_{\text{in}}}, \tag{26}$$

for $c_{\text{in}}\in\{0,1,2\}$ and $k=0,\dots,7$, with the remaining channels in $W_{\text{conv1},c}$ left at zero (so neuron $8c_{\text{in}}+k$ responds only to channel c_{in}). Padding rows $24,\dots,31$ are filled with i.i.d. He noise $\mathcal{N}(0,2/(3\cdot 9))$.

Conv2. For both MNIST and CIFAR, conv2 is square (in/out channels equal) and uses identity-plus-noise on the channel dimension with a single *center-impulse* on each diagonal:

$$W_{\text{conv2},c,c',i,j} = \begin{cases} 1+\eta_{c,c,1,1}, & c'=c,\ (i,j)=(1,1),\\ \eta_{c,c',i,j}, & \text{otherwise,}\end{cases} \tag{27}$$

with $\eta\sim\mathcal{N}(0,\frac{2}{C_{\text{out}}\cdot 9}\cdot 0.05^2)$. That is, the conv2 noise std is $0.05\sqrt{2/(C_{\text{out}}\cdot 9)}$, not 0.01 as in the MLP case.

FC layers. Both fully-connected layers use plain He init: $W_{ij}\sim\mathcal{N}(0,2/n_{\text{in}})$, $b=0$.

C.7. A library of explicit Stage-1 prescriptions

The per-dataset constructions above answer where the theory rows come from for the experiments in this paper. The same question has a general answer: the constructive approximation literature provides explicit weight-level network prescriptions, with sup-norm error bounds, for a wide class of targets (Yarotsky, 2017; Elbrächter et al., 2021). Table 5 collects the prescriptions most relevant as Stage-1 sources. The first block is exact: absolute value, maximum, clamping, the hat function, and one-sided payoffs are ReLU-representable with zero error, so a Stage-1 row drawn from this block carries the target structure exactly (the option payoffs used in the Black–Scholes scenario of Figure 8 are of this form). The second block carries the classical telescoping constructions: squaring via m hat refinements, multiplication via the polarization identity $xy=D^2(u^2-v^2)$ with $u=\lvert x+y\rvert/2D$, $v=\lvert x-y\rvert/2D$, and from these, Horner evaluation of arbitrary polynomials. The third block covers smooth targets: Chebyshev interpolation converts any $f\in C[a,b]$ with analytic continuation into a polynomial with explicitly bounded coefficients, which the Horner construction then realizes; compositions chain the blocks with additive error. We verified the first two blocks numerically: forward passes against the stated bounds for $m=1$ to 5 reproduce the exact identities at zero error, the squaring construction meets $4^{-(m+1)}$ tightly, and the width-6 multiplication meets $D^2\,2^{-2m-1}$ on all sign combinations. The third block carries a practical conditioning limit at fixed depth: converting a Chebyshev interpolant to the monomial basis inflates coefficients as the degree grows, so past a target-dependent degree the chain error from those coefficients outpaces the interpolant improvement (on GELU over $[-3,3]$, no monomial-Horner configuration reaches sup error 0.05 within 40 layers; the frontier is 0.074, reached when the chain converges to its degree-4 interpolant). Basis-aware evaluation of the Chebyshev form is open.

Proof sketch (full details in the cited papers). *Squaring*: truncate the telescoping identity $x^2=x-\sum_{k\ge1} g_k(x)/4^k$, with g_k the k-fold hat composition, after m terms; one refinement layer per term, error $\le 4^{-(m+1)}$ (Yarotsky, 2017, Prop. 2). *Multiplication*: polarization $xy=D^2(u^2-v^2)$ with $u,v=\lvert x\pm y\rvert/2D\in[0,1]$ reduces the product to two squarings; subtracting at the linear output keeps the signed difference out of the ReLU path, error $\le D^2\,2^{-2m-1}$ (Elbrächter et al., 2021, Prop. III.3). *Polynomials*: Horner evaluation, one multiplication stage per degree with the signed partial sum split across two nonnegative channels; the depth bound of Elbrächter et al. (2021, Prop. III.5) grows with $\log\lceil\lVert a\rVert_\infty\rceil$, the coefficient-magnitude cost behind the conditioning limit above. *Compositions*: $h\circ g$ has error $L_h\,\varepsilon_g+\varepsilon_h$, with L_h the Lipschitz constant of h on the range of g.

Endpoints of the prescription spectrum. The library makes the full-prescription endpoint concrete: for the targets above, every weight of every layer can be written down, and two seeds initialized this way are identical at $t=0$ up to any randomized padding, so the symmetry question is resolved trivially ($d_0 \approx 0$) while the function is fixed and nothing is left for training to choose. THEORY-INIT sits at the other end: only the first layer carries content, the $I+\Delta$ middle fixes the frame without fixing the function, and the readout is free. The experiments in this paper show that this minimal prescription already places θ_0 at a canonical orbit representative that survives training; the library is what Stage 1 draws on when a closed-form or smooth target is available, and the general-recipe discussion of Section E.2 covers the case where it is not.

Analytic targets versus domain theory on data. These two settings differ in kind, and the library applies fully only to the first. For an analytic target, the prescription approximates the target function itself: the error is the explicit bound in Table 5, Stage 1 and the target coincide up to that bound, and a fully prescribed network starts at the regression solution up to the bound in Table 5, before seeing any data. On real data, a domain theory (Crippen atomic contributions, Froude scaling, Abrams' law) is a compact but imperfect model of the data-generating process: the prescribed rows encode a useful direction, not the regression function, the residual must be learned, and the accuracy gain depends on how well the theory rows align with the dimensions the trained network uses (Section D.2). The synthetic experiments of the main paper sit between the two: the target is analytic, but the network still fits it from samples with a free readout. Init-time versus trained behavior for both prescription depths is measured next.

Empirical comparison of the two endpoints. We instantiate two library rows as initializations of the paper's architecture and run the standard LMC protocol (4 seeds, 6 within-arm pairs, fixed split, Adam, early stopping): x^2 on $[0,1]$ in $[1,64,64,64,1]$, where the $m{=}3$ squaring construction maps onto $L{=}4$ exactly, and xy on $[-1,1]^2$ in $[2,64,64,64,1]$, where the width-6 multiplication construction fits at $m{=}2$ and a single hidden ReLU layer cannot represent the saddle exactly. Arms: He, orthogonal, the Stage-1 prescription (PWL hinge rows for x^2, the shallow polarization construction for xy, both with the alive-high scaffold), and the full deep prescription (the construction laid down end-to-end, with two padding choices: the small noise of the construction itself, and the $\alpha_1{=}0.1$ padding used elsewhere). Table 7 reports the medians. At $t{=}0$ the full prescription behaves as the library bounds promise: test MSE $8.2{\times}10^{-6}$ on x^2 and $4.5{\times}10^{-5}$ on xy, against $2.0{\times}10^{-4}$ and $1.5{\times}10^{-2}$ for Stage-1. On xy the gap is two to three orders of magnitude: depth buys representation that one hidden layer does not have. After training the ordering reverses. Stage-1 trains to the lowest error on both targets ($1.7{\times}10^{-9}$ on x^2, $1.9{\times}10^{-6}$ on xy) while the full prescription trains to He-level error ($2.6{\times}10^{-7}$, $3.6{\times}10^{-6}$). Stage-1 also holds the canonical frame through training: identity-Hungarian rate 1.00 on both targets, trained normalized distance 0.027 and 0.112, and naive barrier equal to the aligned barrier. The full prescription loses the frame (0.00 identity rate, trained normalized distance 0.56 to 0.62) despite starting nearer ($\hat{d}_0 = 0.05$ to 0.08). Both prescribed arms keep naive barriers two orders of magnitude below the random baselines. The interpretation is again that frame stability tracks the fraction of the first layer that is prescribed, not the init-time accuracy. The deep construction pins 3 to 6 of the 64 first-layer rows and leaves the remaining padding channels free, and training recruits them differently per seed; Stage-1 prescribes most or all of the first layer, so little is left to drift. On these two analytic targets, starting at the regression solution is not the same as training well or staying canonical: the minimal prescription, which fixes the frame and leaves the function to be learned, ends at lower error and a stable frame. These are toy targets by construction (no one needs a network for x^2 or xy); their role is to measure the two ends of the prescription spectrum under controlled conditions, not to claim a general training advantage.

Four further targets, and construction-native depth. Table 6 repeats the measurement on four further library targets: GELU and erf on $[-3,3]$, $e^{-(x^2+y^2)}$ on $[-2,2]^2$, and $x^2{+}y^2$ on $[-1,1]^2$, each at the paper architecture and additionally at the construction-native thin-deep architecture (widths 8–9, depths 4/23/32/38), with all four arms retrained depth-matched. The Stage-1 pattern holds in every cell: identity-Hungarian rate 1.00, naive barrier equal to the aligned barrier, barriers two to five orders below the random arms, and trained error competitive with or below orthogonal init (Stage-1 reaches the best trained error of its cell at depth 32, $4.6{\times}10^{-8}$). Construction-native depth is hostile to random init: orthogonal collapses on half the seeds at depth 32 and on all seeds at depth 38 (trained error equals initial error there, so its barrier entries carry no connectivity information), while both prescription arms train at every depth. The full prescription starts at its verified construction error everywhere, but its trained behavior splits by prescribed fraction. Where the construction occupies most of the width (widths 8–9), the identity survives and $x^2{+}y^2$ at $L{=}4$ records the lowest barrier in the program ($1.4{\times}10^{-7}$); where padding dominates (width 64), the identity dissolves even though training is unaffected; at depth 32 the full prescription is glassy, training two orders worse than Stage-1 ($7.7{\times}10^{-6}$ vs $4.6{\times}10^{-8}$) with a 0.32 barrier between seeds that started at $\hat{d}_0 = 0.002$. Starting orders of magnitude closer at $t{=}0$ guarantees neither better training nor a stable frame; on all six targets tested, the minimal Stage-1 prescription is the robust point of the spectrum.

Table 6. Four further targets at the paper architecture ($[d_{\text{in}}, 64, 64, 64, 1]$) and at the construction-native thin-deep architecture (medians over 6 within-arm pairs, 4 seeds; protocol as in Table 7). The full arm at the paper architecture exists only for x^2+y^2 (its construction has depth 4); the GELU/erf/Gaussian constructions need depths 32/38/23. GELU and erf full arms use their best-frontier configurations (declared bounds 0.080/0.110; see the conditioning note in the text). *orthogonal at depth 38 never leaves its initialization (trained = initial error on all seeds); its barrier entries are between identically untrained networks.

target	arch	arm	t=0 MSE	trained MSE	$\hat{d}_0$	$\hat{d}_{\text{nv}}$	$\mathcal{B}_{\text{nv}}$	id-L1
GELU	paper	He / Orth.	3.6 / 1.5	2.2e-6 / 2.1e-8	1.40	1.40	1.26 / 1.18	0
	paper	Stage-1	2.9e-3	2.7e-8	0.066	0.091	9.3e-5	1.00
	native(32)	Stage-1	1.0e-2	**4.6e-8**	0.041	0.187	4.9e-3	1.00
	native(32)	full	1.9e-3	7.7e-6	0.002	0.084	3.2e-1	0.00
erf	paper	Stage-1	7.2e-3	5.7e-8	0.065	0.097	6.3e-5	1.00
	native(38)	Orth.*	7.3e-1	7.3e-1	1.38	1.38	—	0
	native(38)	Stage-1	6.0e-2	2.0e-7	0.041	0.156	1.1e-2	1.00
	native(38)	full	2.8e-3	6.3e-7	0.000	0.024	2.3e-2	0.17
$e^{-(x^2+y^2)}$	paper	Stage-1	6.6e-4	4.0e-7	0.057	0.060	2.9e-6	1.00
	native(23)	Stage-1	3.7e-3	6.7e-7	0.039	0.302	3.9e-4	1.00
	native(23)	full	5.3e-4	4.9e-6	0.003	0.087	4.0e-3	**1.00**
x^2+y^2	paper	Stage-1	2.3e-2	3.0e-6	0.085	0.093	4.7e-3	1.00
	paper	full	2.9e-5	2.6e-6	0.046	0.067	**1.4e-7**	0.00
	native(4, w=8)	Stage-1	7.2e-3	5.8e-5	0.026	0.325	3.2e-2	1.00
	native(4, w=8)	full	2.9e-5	2.6e-6	0.007	0.157	5.8e-6	1.00

Table 7. Two endpoints of the prescription spectrum (medians over 6 within-arm pairs, 4 seeds; LMC protocol of Section F.1). *Stage-1* is the first-layer prescription with the alive-high scaffold; *full (scaffold)* embeds the m=3 squaring construction with the α_1=0.1 padding used elsewhere; *full (native)* embeds the same construction with its native 0.01-scale padding. $\hat{d}_0$ and $\hat{d}_{\text{nv}}$ are init / trained distances normalized by the pair's mean weight norm (≈ 1.41 for two independent random vectors); id-L1 is the fraction of pairs whose Hungarian-optimal layer-1 (channel) permutation is the identity.

target	arm	t=0 MSE	trained MSE	$\hat{d}_0$	$\hat{d}_{\text{nv}}$	$\mathcal{B}_{\text{nv}}$	id-L1
x^2	He	5.5×10^{-1}	1.9×10^{-7}	1.40	1.40	2.1×10^{-1}	0.00
	Orthogonal	2.1×10^{-1}	2.5×10^{-8}	1.41	1.41	1.4×10^{-1}	0.00
	Stage-1	2.0×10^{-4}	$\mathbf{1.7 \times 10^{-9}}$	**0.000**	**0.027**	$\mathbf{1.1 \times 10^{-4}}$	**1.00**
	full (scaffold)	2.9×10^{-4}	1.5×10^{-8}	0.178	0.503	1.1×10^{-4}	0.33
	full (native)	$\mathbf{8.2 \times 10^{-6}}$	2.6×10^{-7}	0.076	0.615	2.4×10^{-4}	0.00
xy	He	7.8×10^{-1}	2.9×10^{-6}	1.40	1.41	1.1×10^{-1}	0.00
	Orthogonal	1.2×10^{-1}	1.6×10^{-6}	1.41	1.41	1.1×10^{-1}	0.00
	Stage-1	1.5×10^{-2}	$\mathbf{1.9 \times 10^{-6}}$	0.092	**0.112**	$\mathbf{4.2 \times 10^{-4}}$	**1.00**
	full (native)	$\mathbf{4.5 \times 10^{-5}}$	3.6×10^{-6}	**0.053**	0.560	6.4×10^{-4}	0.00

D. Full empirical results

D.1. UCI 9-method comparison

Table 8 reports a 5-dataset $\times$ 9-method comparison run for this submission. 20 seeds per (dataset, method), early stopping with patience 200, max 3,000 epochs, batch size $\min(256, n_{\text{train}})$, Adam ($\beta_1 = 0.9, \beta_2 = 0.999$), learning rate 10^{-3}, float64 precision. Primary metric: median best-validation MSE across 20 seeds (best-validation model restored before test evaluation). THEORY-INIT achieves per-seed Friedman mean rank $1.71/9$, ranking first on 3 of the 5 datasets (Energy, Kin8nm, Yacht).

Table 8. UCI 9-method comparison. Median best-validation MSE (20 seeds), early stopping (patience=200). **Bold** = best per dataset. Rank in parentheses. Friedman mean rank (per-seed) in last column.

Method	Airfoil	Concrete	Yacht	Energy	Kin8nm	Mean Rank
THEORY-INIT (ours)	0.0570 (2)	0.1111 (3)	**0.0016** (1)	**0.0018** (1)	**0.0735** (1)	**1.71**
Orthogonal	**0.0528** (1)	**0.1053** (1)	0.0073 (3)	0.0030 (3)	0.0757 (2)	2.71
GradInit	0.0641 (4)	0.1119 (4)	0.0068 (2)	0.0032 (4)	0.0789 (4)	3.50
Xavier	0.0586 (3)	0.1079 (2)	0.0095 (5)	0.0039 (5)	0.0799 (5)	3.76
ZerO Init	0.1038 (6)	0.1245 (5)	0.0095 (4)	0.0024 (2)	0.0764 (3)	3.92
He	0.1018 (5)	0.1354 (6)	0.0235 (6)	0.0134 (6)	0.1045 (6)	6.04
LSUV	0.1131 (7)	0.1601 (9)	0.0341 (7)	0.0285 (8)	0.1200 (7)	7.26
Data-Dependent	0.1474 (9)	0.1583 (7)	0.0346 (8)	0.0222 (7)	0.1221 (8)	7.70
MetaInit	0.1412 (8)	0.1584 (8)	0.0428 (9)	0.0407 (9)	0.1383 (9)	8.40

D.2. Molecular LogP novelty ablation

To confirm that the gain is specific to *initialization* and not to any use of Crippen domain knowledge, we compare THEORY-INIT against three controls that share the same domain knowledge but inject it differently. 20 seeds per arm; metric is unnormalized RMSE on Molecular LogP held-out test molecules.

Table 9. Novelty ablation on Molecular LogP. All four methods share the same Crippen atomic-contribution information; only THEORY-INIT injects it as initialization. 20 seeds, unnormalized RMSE.

Method	Mechanism	Mean ± std	Median	Δ_{mean}	$\Delta_{\text{med.}}$
THEORY-INIT (ours)	Init weights	$\mathbf{0.747 \pm 0.027}$	$\mathbf{0.739}$	$\mathbf{-7.0\%}$	$\mathbf{-8.2\%}$
Frozen (N3)	Freeze theory layer	0.796 ± 0.034	0.795	-0.9%	-1.2%
Feature aug. (N2)	Theory as input feature	0.799 ± 0.024	0.800	-0.5%	-0.5%
Residual (N1)	Theory + He correction	0.807 ± 0.037	0.801	$+0.5\%$	-0.4%
He (baseline)	Random init	0.803 ± 0.035	0.805	0.0%	0.0%

Only THEORY-INIT achieves a large effect (-8.2% median). The frozen, feature-augmented, and residual variants all land within $\pm 1.3\%$ of He. The gain comes from the initialization itself, not from feeding the theory in as a feature, a residual, or a frozen layer.

D.3. Full LMC results: per-dataset, per-pair-type

Table 10 extends the main paper Table 1 from 3 datasets to all 4 (adds Concrete) and adds the cross-scheme He–THEORY-INIT pair type. Aggregated medians over $\{6, 6, 6, 16\}$ pairs respectively. $\mathcal{B}_{\text{nv}}, \mathcal{B}_{\text{al}}$ are naive and Git Re-Basin-aligned barriers (max excess loss along the linear interpolation, $\alpha \in \{0, 0.05, \ldots, 1\}$). $d_{\text{nv}}, d_{\text{al}}$ are pre/post-alignment Frobenius distances. All medians are computed over the within-scheme pairs of the four-seed runs.

Table 10. Full LMC results across 4 datasets and 4 pair types. Medians over within-scheme pairs ($n = 6$ each, $n = 16$ for cross-scheme He–Theory). Lower is better (closer pre-alignment, smaller barrier). $\hat{d}_{\text{nv}} = d_{\text{nv}}/\|\theta\|_F$ is the scale-free naive distance (trained-time Frobenius distance divided by the pair's reference weight norm); it shows that THEORY-INIT's smaller distance is *not* merely an artifact of its smaller weight mass.

Dataset	Pair	n	$\mathcal{B}_{\text{nv}}$	$\mathcal{B}_{\text{al}}$	d_{nv}	d_{al}	$\hat{d}_{\text{nv}}$
Yacht	He–He	6	1.197	0.427	28.81	22.14	1.42
	Ortho–Ortho	6	1.030	0.106	18.18	14.82	1.41
	THEORY-INIT–THEORY-INIT	6	**0.288**	**0.203**	**8.14**	**7.99**	**0.54**
	He–THEORY-INIT (cross)	16	1.186	0.533	25.59	22.02	1.26
Energy	He–He	6	0.844	0.629	29.21	22.87	1.42
	Ortho–Ortho	6	0.800	0.330	18.50	15.50	1.42
	THEORY-INIT–THEORY-INIT	6	**0.553**	**0.511**	**10.31**	**10.18**	**0.66**
	He–THEORY-INIT (cross)	16	0.889	0.438	25.62	22.34	1.25
Concrete	He–He	6	1.114	0.823	31.60	25.00	1.40
	Ortho–Ortho	6	0.823	0.440	22.28	17.18	1.32
	THEORY-INIT–THEORY-INIT	6	**0.605**	**0.476**	**13.93**	**13.92**	**0.83**
	He–THEORY-INIT (cross)	16	1.761	4.162	27.49	23.58	1.23
LogP	He–He	6	0.806	1.043	29.90	25.78	1.40
	Ortho–Ortho	6	0.532	0.690	24.38	19.75	1.44
	THEORY-INIT–THEORY-INIT	6	**0.477**	**0.408**	**16.39**	**15.92**	**1.04**
	He–THEORY-INIT (cross)	16	0.694	0.862	27.19	23.55	1.28

Summary. THEORY-INIT–THEORY-INIT pairs have the smallest *naive* barrier $\mathcal{B}_{\text{nv}}$ and the smallest pre-alignment distance on every dataset. The *aligned* barrier $\mathcal{B}_{\text{al}}$, by contrast, is higher for THEORY-INIT than for Ortho on $3/4$ datasets (Yacht $0.20 > 0.11$, Energy $0.51 > 0.33$, Concrete $0.48 > 0.44$; LogP is the exception): Ortho retains full permutation symmetry and the matching search removes a large barrier to reach a low floor, whereas THEORY-INIT removed that symmetry by construction so the search has almost nothing left to exploit. The naive barrier is the metric matched to our claim (closeness before any alignment); the aligned barrier mainly measures the residual symmetry a scheme left behind. The aligned-vs-naive distance reduction $d_{\text{nv}} - d_{\text{al}}$ is negligible for THEORY-INIT–THEORY-INIT (0.01–0.50 absolute) but 5–8 for He–He, indicating identity is the optimal permutation already for THEORY-INIT. The proximity is not a smaller-weight-mass artifact: the scale-free naive distance $\hat{d}_{\text{nv}} = d_{\text{nv}}/\|\theta\|_F$ is still 0.54–1.04 for THEORY-INIT versus 1.40–1.44 for He on every dataset (e.g. Yacht 0.54 vs 1.42, a $\sim 2.6\times$ separation after dividing out weight mass), so deterministic placement, not merely a smaller weight norm, drives the smaller distance. On LogP and Concrete, layer-wise weight matching is unstable in the He–theory cross-scheme regime (aligned barrier exceeds naive), consistent with prior reports that neuron-alignment methods alone are insufficient for low-barrier interpolation in low-overparameterization settings (Ainsworth et al., 2023; Jordan et al., 2023).

D.4. Init-time distance and scrambled-physics ablation

Table 11 reports the init-time Frobenius distance $d_0 = \|\theta_0^{(A)} - \theta_0^{(B)}\|_F$ for all $\binom{4}{2} = 6$ within-scheme pairs at $t = 0$ (no training). THEORY-INIT's tight init-time distance (~ 2 on UCI tasks, ~ 5 on LogP) is an order of magnitude below He's (~ 28 on all datasets).

Table 11. Init-time distance $d_0 = \|\theta_0^{(A)} - \theta_0^{(B)}\|_F$ across 6 within-scheme pairs. Median (min–max) over the 6 pairs.

Dataset	He–He	Ortho–Ortho	THEORY-INIT–THEORY-INIT
Yacht	27.81 (27.44–28.42)	16.35 (16.25–16.45)	**1.96 (1.93–2.00)**
Energy	27.93 (27.43–28.30)	16.46 (16.35–16.57)	**2.04 (2.01–2.05)**
Concrete	27.93 (27.43–28.30)	16.46 (16.35–16.57)	**2.07 (2.07–2.14)**
LogP	27.68 (27.43–27.90)	19.52 (19.38–19.67)	**4.86 (4.82–4.92)**

Scrambled-physics ablation. To rule out "LSAP returns I because the cost matrix is constructed to be diagonal", we randomly permute the first N rows of θ_B's layer-1 (the physics neurons), breaking the prescribed Stage-2 row assignment, and re-run weight matching. Table 12 shows that the algorithm correctly recovers the inverse scramble: distance returns to its baseline value after the recovery permutation is applied. This confirms that the no-op finding is a property of the prescribed Stage-2 row assignment, not a cost-matrix tautology.

Table 12. Scrambled-physics ablation: randomly permuting the first N rows of θ_B's layer-1 increases distance, but weight matching recovers the inverse scramble. Medians over 4 THEORY-INIT–THEORY-INIT pairs per dataset.

Dataset	d_{baseline}	$d_{\text{scrambled}}$	$d_{\text{recovered}}$
Yacht	8.44	14.08	8.05
Energy	10.32	14.06	10.27
LogP	15.92	16.39	16.13

The scrambled-vs-recovered distance swing is 1.67–1.75$\times$ on Yacht and 1.36–1.40$\times$ on Energy ($8.44 \to 14.08 \to 8.05$, $10.32 \to 14.06 \to 10.27$); the LogP swing is smaller because LogP has more non-physics noise overall ($d = 64$ with $\alpha_1 = 0.1$ padding noise of substantial magnitude). The recovery succeeds in all 12 ablation runs.

D.5. Cross-scheme He–theory pairs

The He–THEORY-INIT cross-scheme pair type (last row in each block of Table 10) is the natural "can we LMC across initializations" question: for each (dataset) we have 4 He nets and 4 THEORY-INIT nets, giving $4 \times 4 = 16$ cross-pairs. The cross-pair distance $d_0 \approx 28$ is similar to He–He, as expected (a He init and a THEORY-INIT init differ by $\sim$ Frobenius norm of a He init, since the THEORY-INIT physics rows account for a small fraction of the total weight mass). Aligned barriers: on Yacht/Energy, the alignment helps ($1.19 \to 0.53$, $0.89 \to 0.44$), but on Concrete/LogP it does not (aligned barrier exceeds naive). This is consistent with the layer-wise instability for cross-scheme LMC: the optimization landscapes around a He minimum and a THEORY-INIT minimum need not be permutation-equivalent at the layer level.

D.6. Extended ablations

We present five additional ablations characterizing when the pre-alignment effect occurs. The corresponding paragraph in the main paper ("Scale of the barriers and further evidence", § 4) summarizes (i)–(iii) below; (iv) and (v) appear only here.

(i) Within-scheme regression $d_{\text{nv}} \sim$ train_mse$+\mathbb{1}\{$theory$\}$. An alternative reading of the small THEORY-INIT distances is that "better-fit nets live in a smaller submanifold". We test this by fitting an OLS regression on $n = 54$ within-scheme pairs (3 schemes $\times$ 6 pairs $\times$ 3 datasets):

$$d_{\text{nv}} = 23.0 + 11.9\,\text{train\_mse} - 13.0\,\mathbb{1}\{\text{theory}\}, \quad R^2 = 0.74. \tag{28}$$

The theory dummy carries $t = -11.3$ ($p < 10^{-14}$, robust SE): the symmetry-breaking effect persists at -13 Frobenius units after controlling for train-MSE. Per-method correlations: $\rho(\text{train\_mse}, d_{\text{nv}}) = 0.81$ (He), 0.95 (THEORY-INIT), 0.99 (orthogonal).

(ii) σ_Δ sweep on Yacht (5 values $\times$ 4 seeds). Varying the middle-layer noise scale $\sigma_\Delta \in \{0, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$:

σ_Δ	0	10^{-3}	10^{-2}	10^{-1}	1
d_{nv} med	8.2	8.2	8.5	15.7	126.8
$\mathcal{B}_{\text{nv}}$ med	0.30	0.29	0.24	0.25	2.71
identity-opt	$0/6$	$1/6$	$3/6$	$4/6$	$0/6$

The signature is preserved across $\sigma_\Delta \in [0, 10^{-1}]$ (distance $\leq 55\%$ of the He-He median 28.8); only at σ_Δ=1, where the middle layers are no longer near identity, does the effect collapse. It is the identity *anchor*, not the magnitude of the perturbation, that drives the small-distance, identity-optimal LMC signature.

(iii) N-scan on LogP ($N \in \{0, 1, 2, 4, 8, 16, 32, 64\}$, 4 seeds each). Varying the number of physics neurons in layer 1, with the remaining $w - N$ rows He-random:

N	0	1	2	4	8	16	32	64
d_{nv} med	10.8	12.5	12.4	12.5	13.0	13.6	10.4	13.1
identity-opt	$0/6$	$0/6$	$3/6$	$5/6$	$3/6$	$4/6$	$6/6$	$1/6$

The naive distance is roughly N-independent (the small-distance signature is partly an $I + \Delta$ effect), but *identity-Hungarian-optimality requires theory rows*: $0/6$ at N=0, climbing to $5/6$ at N=4 and $6/6$ at N=32. Equivalently: $I + \Delta$ alone yields small distance but *not* the canonical Hungarian assignment; both require Stage 2's index assignment.

(iv) Tiny-He magnitude baseline (Yacht, $\beta \in \{0.1, 0.2, 0.5, 1.0\}$). Scaled-down He init ($\beta$=0.1) starts at d_0=2.8 (close to THEORY-INIT's 1.96), but its post-training distance is 10.3 vs. THEORY-INIT's 8.1, and Git Re-Basin alignment finds much lower aligned barriers ($\mathcal{B}_{\text{nv}}$=1.05 $\to$ $\mathcal{B}_{\text{al}}$=0.02, a $\sim$50$\times$ drop) unlike THEORY-INIT (0.29 $\to$ 0.20, 1.5$\times$). Tiny-He has small init distance *without* prescribed Stage-2 row assignment.

(v) Frozen-theory comparison (Yacht). Training the same THEORY-INIT network with the first layer *frozen* gives d_{nv}=11.2 (median, 6 pairs) and identity-optimal in $4/6$ pairs, larger than the trainable case (d_{nv}=8.1, identity-optimal in $5/6$). A direct checkpoint-level decomposition (Table 13) explains why: with the first layer L_0 anchored, the hidden layers L_1 and L_2 absorb the structure L_0 would otherwise carry, drifting further from init and pulling apart more across seeds. The full-parameter inflation (11.2$-$8.1=3.1) is concentrated in L_2 (+2.76) and L_1 (+1.71). The trainable run, by contrast, spreads adaptation across all four weight matrices and yields a more coordinated, lower-variance solution manifold across seeds. The cross-pair distance between a frozen-theory net and a trainable theory net (matched seed) is 9.8, between the two within-condition medians, consistent with this account (Figure 5).

Table 13. Per-layer decomposition, frozen vs. trainable THEORY-INIT on Yacht (layers L_0–L_3). Drift is $\|W_{\text{final}} - W_{\text{init}}\|_F$; cross-seed distance is the trained-time pairwise Frobenius distance. Freezing L_0 shifts both quantities into the hidden layers.

Quantity	Condition	L_0	L_1	L_2	L_3
Drift	trainable	1.04	3.98	3.87	0.38
	frozen	0.00	5.01	5.80	0.56
Cross-seed	trainable	2.04	5.53	5.57	0.63
	frozen	1.49	7.24	8.32	0.89

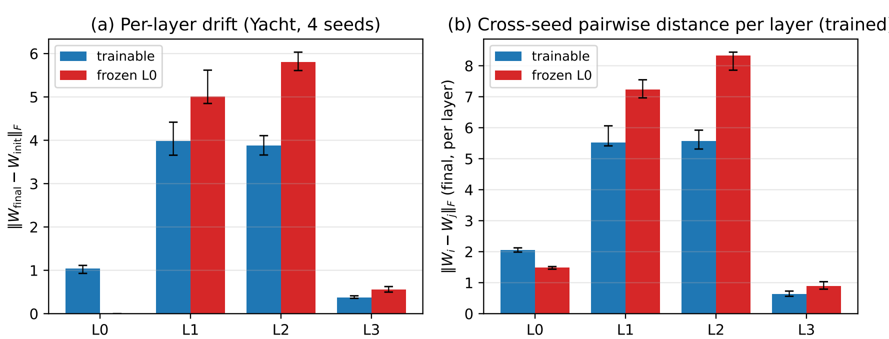

Figure 5. Per-layer drift and cross-seed distance, frozen vs. trainable THEORY-INIT on Yacht. With L_0 frozen, L_1 and L_2 absorb the dataset-specific structure, and the cross-seed inflation is concentrated in L_2 rather than directly downstream of the anchor.

Functional pre-alignment (post-hoc). The midpoint-function distance $\|f((\theta_A + \theta_B)/2) - \frac{1}{2}[f(\theta_A) + f(\theta_B)]\|^2$ on test inputs is uniformly smaller for THEORY-INIT than for He (Yacht 0.286 vs. 1.120; Energy 0.538 vs. 0.797; LogP 0.523 vs. 0.746; Concrete 0.598 vs. 1.189). The pre-alignment is functional, not just parametric.

Cross-seed weight similarity. The cross-seed similarity numbers in Table 10 and the per-layer drift in Figure 5 are weight-space statistics; Figure 6 visualizes them directly. Two seeds of He training produce nearly orthogonal layer-0 weight matrices ($\bar{c}_{\text{off}}\approx -0.01$) while two seeds of THEORY-INIT training produce nearly identical layer-0 matrices ($\bar{c}_{\text{off}}\approx 0.97$, panel A), driven by the four deterministic physics rows that retain $\sim 20\times$ larger magnitude than the He-padding rows even after ~ 2.6k training epochs (panel B).

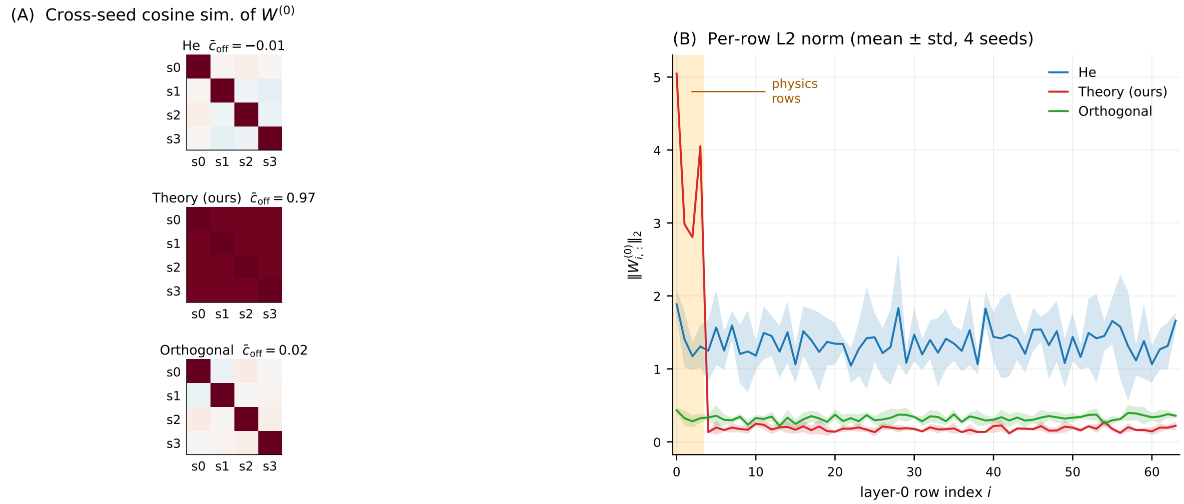

Figure 6. **THEORY-INIT networks share an orbit representative across seeds, at t=0 and after training.** Yacht Hydrodynamics (d_{in}=6, w=64), four checkpoints per init. **(A)** Cross-seed cosine similarity of vectorized $W^{(0)}$: off-diagonal mean $\bar{c}_{\text{off}}$=0.97 for Theory vs. -0.01 (He) and 0.02 (Orthogonal), so THEORY-INIT-trained seeds remain nearly co-linear in layer-0 while He / Orthogonal pairs are nearly orthogonal. **(B)** Per-row L2 norm of trained $W^{(0)}$, mean $\pm$ std over four seeds: the first four theory rows have $\|W^{(0)}_{i,:}\|_2 \approx 3.7$ vs. ≈ 0.18 for the He-padded rows ($\sim 20\times$), a sparse physics block on top of a He-random body. He and Orthogonal show no row-wise structure.

D.7. Scale-up: small CNN on MNIST and CIFAR-10

To extend beyond MLPs and regression we test whether the LMC pre-alignment effect replicates on small CNNs. **Architecture:** `Conv(`$c_{\text{in}}\to f$`, 3×3)`–ReLU–MaxPool$_2$–`Conv(`$f\to f$`, 3×3)`–ReLU–MaxPool$_2$–`FC(`$f{\cdot}s^2\to h$`)`–ReLU–`FC(`$h\to 10$`)`. For MNIST: c_{in}=1, f=16, h=64, s=7 ($\sim$53k params). For CIFAR-10: c_{in}=3, f=32, h=128, s=8 ($\sim$270k params).

THEORY-INIT for CNN: the first N=8 output channels of the first convolution are hand-crafted physics filters (Sobel-X/Y, Laplacian, zero-mean Gaussian, oriented DoGs at $0/45/90/135°$), each averaged over input channels and normalized to He-magnitude; remaining channels are He-random. The second convolution uses a center-impulse identity (the analog of $I+\Delta$ for square channel maps): $W_{c,c,1,1} \mathrel{+}= 1$ plus small Gaussian noise (σ=0.05). Both fully-connected layers use He-random (no natural identity analog for non-square layers). **Training:** cross-entropy, Adam at lr $= 10^{-3}$, batch size 128, 15 epochs (MNIST) or 30 (CIFAR) with early stopping (patience 5 / 8 on validation loss); 80/20 train/validation split with a fixed data seed shared across paired networks. Final test accuracies (over 4 seeds $\times$ 3 inits): MNIST 98.6–99.0% (matches LeNet baselines), CIFAR-10 66.6–69.9% (typical for a small unaugmented CNN at this depth).

Conv-aware weight matching. The Ainsworth procedure handles three permutations: P_{c1} (size f, first-conv output), P_{c2} (size f, second-conv output), P_{f1} (size h, FC1 output). For FC1's input we treat the flattened $f{\cdot}s^2$ vector as f blocks of s^2 spatial positions and apply P_{c2} to the channel axis; the classifier output has fixed indices.

Table 14. LMC on small CNNs (medians over 6 within-scheme pairs).

Dataset	Pair	$\mathcal{B}_{\text{nv}}$	$\mathcal{B}_{\text{al}}$	d_{nv}	P^*=I
MNIST	He–He	1.56	0.13	25.6	0/6
	Ortho–Ortho	1.22	0.03	20.4	0/6
	THEORY-INIT–THEORY-INIT	**0.72**	**0.08**	**24.0**	0/6
CIFAR-10	He–He	1.24	0.61	35.8	0/6
	Ortho–Ortho	1.15	0.13	35.1	0/6
	THEORY-INIT–THEORY-INIT	**1.12**	**0.73**	34.2	0/6

On MNIST, the pre-alignment asymmetry replicates: THEORY-INIT–THEORY-INIT's 0.72 is 46% of He–He's 1.56, within

the 24–66% range observed on the four small-MLP datasets. After Git Re-Basin alignment, all three schemes achieve $\mathcal{B}_{\text{al}} \leq 0.13$, so LMC does hold post-alignment, consistent with classical CIFAR/CNN findings (Ainsworth et al., 2023).
On CIFAR-10 with channel-averaged physics filters (8 filters, each averaged across the three RGB input channels, the direct analog of the MNIST construction), the asymmetry weakens: THEORY-INIT–THEORY-INIT naive barrier (1.12) is 90% of He–He's (1.24), at the upper edge of the small-MLP range. We attribute this to channel averaging discarding the color information that the trained network uses, so the layer-1 physics neurons under-specify the prescribed Stage-2 row assignment.
A *per-channel* variant tests this: 24 physics filters (8 physics directions $\times\, 3$ RGB), where output channel $8c + k$ ($c \in \{R, G, B\}$, $k \in \{0, \ldots, 7\}$) responds to physics filter k applied to color channel c only. The remaining 8 output channels are He padding. With the same architecture, hyperparameters, and training protocol, the THEORY-INIT–THEORY-INIT naive barrier drops to 62% of He–He's, back inside the small-MLP range (Table 15). The pre-alignment ratio is robust to the constant per-channel rescaling of filter magnitude. Final test accuracy stays in the 68–71% range across both variants.

Table 15. CIFAR-10 per-channel THEORY-INIT: 24 physics filters (3 RGB $\times$ 8 physics) recovers the pre-alignment effect lost in the averaged variant.

Pair	$\mathcal{B}_{\text{nv}}$	$\mathcal{B}_{\text{al}}$	d_{nv}	vs He
He–He	1.24	0.61	35.8	–
Ortho–Ortho	1.15	0.13	35.1	93%
THEORY-INIT (avg)	1.12	0.73	34.2	90%
THEORY-INIT (p-c)	**0.77**	0.29	34.5	**62%**

The per-channel CIFAR result indicates that the size of the pre-alignment effect tracks physics-filter coverage of input axes, not architectural scale. On grey-scale (MNIST) or low-dimensional tabular (UCI / MoleculeNet) inputs, the standard 8 physics filters cover most of the relevant axes; on RGB inputs the filters must be replicated per channel for comparable coverage.

Scope across settings. Across the four small-MLP UCI / MoleculeNet datasets, MNIST CNN, and the two CIFAR-10 CNN configurations, $\mathcal{B}^{\text{TH}}_{\text{nv}} / \mathcal{B}^{\text{He}}_{\text{nv}} < 1$ in every case, with ratios 24–66% (small-MLP UCI / MoleculeNet), 46% (MNIST CNN), 90% (CIFAR-10 with channel-averaged physics filters), and 62% (CIFAR-10 with per-channel physics filters; see Section D.7 and Table 15). The averaged-versus-per-channel comparison on CIFAR-10 shows that the apparent CIFAR weakness is driven by under-specification of the physics-neuron set when input channels carry independent signal, not by architectural scale.

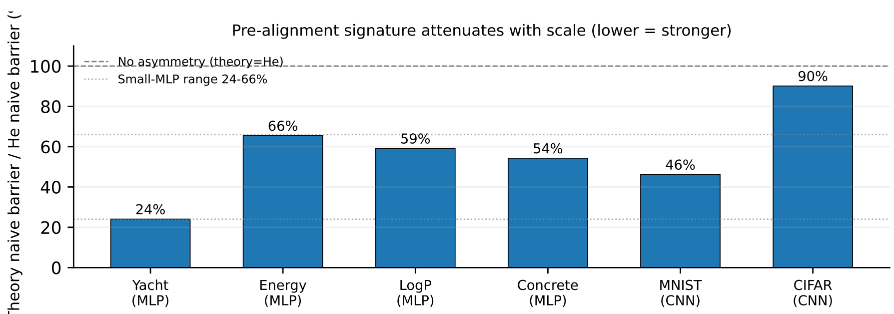

Figure 7. Naive-barrier ratio $\mathcal{B}^{\text{TH}}_{\text{nv}} / \mathcal{B}^{\text{He}}_{\text{nv}}$ across the four small-MLP datasets and two CNN scale-ups (lower = stronger). The bar shown for CIFAR-10 is the channel-averaged variant (90%), which under-covers the RGB axes. The per-channel CIFAR-10 variant (62%, Table 15, not plotted here) sits inside the small-MLP range, so the effect tracks physics-filter coverage rather than parameter count; the figure title overstates the scale story.

Depth and residual variant for the per-channel prescription. The per-channel CIFAR-10 prescription of Table 15 uses two square convolutions above the prescribed conv1. To test whether the symmetry-breaking signature transmits through more depth and through a residual stream, we vary only the depth above conv1 in $\{2, 3, 4, 5\}$ plus a single residual variant at depth 4 (two residual blocks of two convs each, one skip per block, the residual stream sharing channel identity with conv1's output). Everything else is held fixed: CIFAR-10 split (data seed 999), width 32, optimizer, and matcher. Four init schemes per configuration: He, orthogonal, THEORY-INIT (the per-channel prescription of Section C.6), and THEORY-INIT-*scrambled*, a control that permutes the 24 prescribed filters across the 24 conv1 slots per seed, preserving the magnitude profile and destroying the binding of filter role to slot. Four seeds per cell; six within-scheme pairs. Total 80 trainings. Sanity. A random valid channel permutation applied to a trained network preserves the network function to float32 precision (max $|\Delta| = 4.9 \times 10^{-4}$ across all five configurations); self-match returns the identity in every configuration.

Table 16. CNN depth + residual pilot (CIFAR-10, width 32, 4 seeds per cell, 6 within-scheme pairs). THEORY-INIT is the per-channel CIFAR prescription of Table 15 ($24 = 3\,\text{RGB} \times 8$ physics on conv1, identity-plus-noise on subsequent square convs, He on FC); THEORY-INIT-*scrambled* permutes those 24 filters across conv1 slots per seed, preserving magnitudes but destroying the binding of filter role to slot position. Medians; $d_{\text{norm}} = 1.41$ is the random-vector baseline. id_opt is the fraction of pairs whose Hungarian-optimal channel permutation is the identity. Source: Section F.1, 80 CIFAR-10 trainings.

depth / variant	scheme	$\mathcal{B}_{\text{nv}}$	d_{norm}	id_opt	acc
$D = 2$ plain	He	1.31	1.41	$0/6$	0.68
	Orthogonal	1.32	1.41	$0/6$	0.69
	THEORY-INIT	$\mathbf{0.82}$	$\mathbf{1.32}$	$0/6$	0.71
	THEORY-INIT-scrambled	1.16	1.34	$0/6$	0.71
$D = 3$ plain	He	1.35	1.41	$0/6$	0.68
	Orthogonal	1.32	1.41	$0/6$	0.66
	THEORY-INIT	$\mathbf{0.86}$	$\mathbf{1.28}$	$0/6$	0.71
	THEORY-INIT-scrambled	1.18	1.31	$0/6$	0.72
$D = 4$ plain	He	1.35	1.41	$0/6$	0.68
	Orthogonal*	0.34	1.62	$0/6$	0.24
	THEORY-INIT	$\mathbf{0.94}$	$\mathbf{1.26}$	$0/6$	0.72
	THEORY-INIT-scrambled	1.22	1.27	$0/6$	0.71
$D = 5$ plain	He	1.39	1.41	$0/6$	0.68
	Orthogonal*	0.66	1.83	$0/6$	0.38
	THEORY-INIT	$\mathbf{0.96}$	$\mathbf{1.22}$	$0/6$	0.71
	THEORY-INIT-scrambled	1.24	1.25	$0/6$	0.71
$D = 4$ residual	He	1.31	1.41	$0/6$	0.68
	Orthogonal	1.29	1.41	$0/6$	0.72
	THEORY-INIT	$\mathbf{0.86}$	$\mathbf{1.22}$	$0/6$	0.71
	THEORY-INIT-scrambled	1.19	1.24	$0/6$	0.71

*Orthogonal collapses during training at $D = 4$ and $D = 5$ plain (acc 0.24 and 0.38); LMC numbers for those two cells reflect under-trained networks. The residual variant restores Orthogonal (acc 0.72), confirming this is an optimization artifact of deep plain conv stacks rather than a symmetry-breaking signal.

Prescribed conv1 slots load at $\|\cdot\|_2 = \sqrt{2/9}$ exactly, as expected. The scale-normalized distance d_{norm} of THEORY-INIT stays below the random-vector baseline $\sqrt{2} \approx 1.41$ at every depth (1.32, 1.28, 1.26, 1.22 plain at $D = 2, 3, 4, 5$, and 1.22 in the residual variant) and stays below the scrambled control at every depth (gaps $0.014, 0.027, 0.007, 0.023$ plain; 0.020 residual; Table 16). The naive barrier $\mathcal{B}_{\text{nv}}$ of THEORY-INIT is below both He and scrambled at every depth. The signature is inherited from conv1 through identity-plus-noise convs and is not specific to a single conv above conv1, and a residual stream does not collapse it. id_opt is $0/6$ in every cell of Table 16, including $D = 2$, matching the original per-channel CIFAR baseline of Table 15. Strict $P^*{=}I$ was never present in this CIFAR configuration and the depth+residual sweep confirms this is depth-invariant, not a deeper-CNN regression. The matcher sanity above rules out an alignment bug. Orthogonal init collapses during training at $D = 4$ and $D = 5$ plain (test accuracy 0.24 and 0.38, against 0.68 for He at the same depths); the residual variant restores it (0.72). The collapse is an optimization artifact of deep plain conv stacks rather than a symmetry-breaking signal, and Orthogonal's LMC numbers in those two cells should be read in that light. THEORY-INIT and He train normally at every depth.

D.8. Breadth across depth and width

To test whether the symmetry-breaking signature is specific to the $[d_{\text{in}}, 64, 64, 64, 1]$ architecture of Table 10, we re-run the within-scheme LMC protocol across four hidden configurations on all four datasets, varying only the hidden-width specification. The configurations are $32{\times}32{\times}32$ (narrower), $128{\times}128{\times}128$ (wider), $64{\times}64$ (two hidden layers), and $64{\times}64{\times}64{\times}64{\times}64$ (five hidden layers), with 3 methods and 4 seeds each (6 within-scheme pairs per cell). Alongside the raw distances we report *scale-normalized* distances $\hat{d} = d/[\frac{1}{2}(\|\theta_A\| + \|\theta_B\|)]$, which removes the magnitude confound: two independent random parameter vectors sit at $\hat{d} \approx \sqrt{2} \approx 1.41$ regardless of their norm.

Table 17. Breadth across depth/width: median over 96 within-scheme pairs per method (16 configs $\times$ 6 pairs). $\hat{d}_0, \hat{d}_{\text{nv}}$ are scale-normalized init / trained distances; Δ is the naive–aligned barrier gap; id-frac is the fraction of pairs whose layer-1 matching permutation is identity.

Method	$\hat{d}_0$	$\hat{d}_{\text{nv}}$	$\mathcal{B}_{\text{nv}}$	Δ	id-frac
THEORY-INIT	$\mathbf{0.181}$	$\mathbf{0.728}$	$\mathbf{0.461}$	$\mathbf{+0.000}$	0.094
He	1.415	1.415	0.970	$+0.255$	0.016
Ortho	1.409	1.412	0.816	$+0.384$	0.021

The patterns hold at every depth and width tested. THEORY-INIT has the lowest init distance and the lowest naive barrier in nearly all 16 configurations. The signature is not a magnitude artifact: after scale normalization, He and orthogonal sit at ≈ 1.41 (the expected normalized distance between two independent random vectors), while THEORY-INIT is 0.181 at init and 0.728 after training, a large reduction that survives normalization. The identity-Hungarian property *strengthens with depth*: the fraction of THEORY-INIT pairs for which weight matching returns the exact identity grows from a few of 24 in the 2–3 hidden-layer nets to $22/24$ in the five-hidden-layer nets (mean id-frac 0.92), across all four datasets. Deeper near-identity middle blocks propagate the canonical frame more faithfully, consistent with the index-preservation argument of Section A.8.

As in Table 10, the *aligned* barrier of THEORY-INIT is higher than the aligned barrier of orthogonal init in 12 of 16 configurations. THEORY-INIT already sits in a canonical permutation frame, so Git Re-Basin has nothing to remove; orthogonal init starts far from the identity frame and the search reduces its aligned barrier below THEORY-INIT's already-low naive barrier. The contribution is "no alignment needed," not "lowest achievable aligned barrier."

E. Additional Discussion

We position THEORY-INIT relative to the closest architecture-side line of work (asymmetric networks, Lim et al., 2024), discuss where the construction is expected to apply beyond the small-MLP regime and what breaks first, and treat the continuous symmetries the head-permutation pilot does not touch.

E.1. Comparison with Asymmetric Networks (Lim et al., 2024)

Lim et al. (2024) give the canonical large-scale empirical study of what parameter symmetries do, by *removing* them and comparing. They build two architecture families: *W-Asymmetric* networks freeze a subset of entries per layer to fixed random constants ($W' = M \odot W + (1 - M) \odot \mathbf{F}$, with $\mathbf{F}$ much larger in magnitude than the trained weights), giving provably no nontrivial permutation symmetries when each mask has unique rows; and *σ-Asymmetric* networks replace the elementwise nonlinearity with a fixed-mixing gated unit (FIGLU, $\sigma(x) = \eta(\mathbf{F}x) \odot x$) that removes permutation *and* diagonal/scaling equivariances. Trained with ordinary optimizers, these show alignment-free linear mode connectivity, Bayesian networks that train faster and at greater depth, more predictable metanetworks, and near-universal monotonic interpolation, across MLPs, ResNets (to $8\times$ width), and graph networks on MNIST/CIFAR/ogbn-arXiv.

The relationship to THEORY-INIT is best read as a duality between acting on the group and acting on the point.

- **What is modified.** They modify the function class f_θ (frozen entries or a non-elementwise nonlinearity), so the symmetric counterpart never exists. We modify only the initialization θ_0 of a standard ReLU MLP and keep the full permutation group intact, selecting one orbit representative at $t{=}0$.
- **Persistence.** Their asymmetry is structural and permanent. Our canonical frame is preserved *in practice* through training (Git Re-Basin returns $\approx I$ between two THEORY-INIT nets, Table 17) but is not enforced by construction.
- **Content.** Their fixed entries are random and content-free, and they concede the large magnitude is a confound for the landscape effects. Our prescribed rows are interpretable hinge / atomic-contribution / scaling functions at *small* magnitude, so they fix the gauge *and*, when aligned with the task, add accuracy (Section D.2). Our own controls show the symmetry-breaking *signature* is itself largely a low-magnitude scaffold effect (Section G.1), the magnitude/placement split is characterized in Section G.1.
- **Trainable parameters.** W-Asymmetric nets have *fewer* trainable parameters (some entries frozen); THEORY-INIT keeps the standard count (nothing is frozen, the theory rows are trainable and merely initialized deterministically).

Two of their results bear directly on our design. Their negative result that *fixing biases fails to break permutation symmetry* (their App. E.3) is independent evidence that *where* the asymmetry is injected matters; our lever is deterministic first-layer placement carried through depth by near-identity layers, not biases. And their distance-versus-interpolation decoupling (well-interpolating asymmetric nets are not necessarily closer in parameter space) contrasts with our pattern, where THEORY-INIT pairs are both closer and lower-barrier (Table 10); the difference reflects canonical-frame placement versus a random fixed scaffold, not a contradiction.

Their study spans ResNet-$8\times$, a real citation graph, and 20k trained models; ours covers four tabular regressions plus a small CNN, with transformer and LoRA as pilots (Section H). We use Lim et al. (2024) as the precedent that *reducing parameter symmetry is consequential beyond small MLPs*, not as a benchmark we set out to outperform. THEORY-INIT contributes a different, strictly lighter intervention: init-time canonical-representative selection on an unchanged architecture, with a domain prior that can also carry accuracy.

E.2. When and how at larger scale, and what breaks first

Lim et al. (2024) establishes that reducing parameter symmetry matters at CIFAR and citation-graph scale; our scale limitation concerns the coverage of the init-time variant, not the premise. We distinguish two regimes for where THEORY-INIT is expected to apply.

Canonical-domain regime (strongest case). When a compact, established model of the target is available in closed form (a Crippen atomic-contribution formula, a Froude-number scaling, an Abrams law), Stage 1 has a principled source for the prescribed rows, and the accuracy bonus appears where the prior aligns with the task (Section D.2). Explicit weight-level prescriptions with error bounds for common closed-form targets, from exact ReLU identities to Chebyshev interpolants of analytic functions, are collected in Section C.7. The symmetry-breaking signature, by contrast, needs only *some* deterministic, magnitude-matched first-layer prescription, so it is available even when no domain theory exists.

General-recipe regime. Absent a domain theory, the scaffold can be made data-driven: deterministic first-layer directions (e.g. leading PCA, or random directions sampled once and shared across seeds) at small magnitude fix the gauge without any task prior, trading the accuracy bonus for the symmetry-breaking signature alone. This is the form we expect to transfer most readily to new architectures.

Prescribed-fraction principle (what governs the effect size). The effect tracks the fraction of the first layer that is prescribed and the coverage of the relevant input axes, not the raw parameter count. The CNN results make this concrete: channel-averaged physics filters on CIFAR-10 leave the color axis under-specified and the effect attenuates to 90% of He, whereas per-channel filters that cover RGB recover it to 62% (Tables 14 and 15). The breadth sweep adds the complementary depth finding: the identity-Hungarian property strengthens with depth (Table 17), so deeper near-identity stacks propagate the canonical frame more, not less, faithfully.

What breaks first. (i) *Continuous symmetries.* The construction targets the discrete permutation group; ReLU positive-scaling and (in attention) rotational $Q/K/V/O$ subspace symmetries remain unfixed and would need a separate canonicalization (Section E.3). (ii) *Padding fraction.* As width grows for a fixed theory-row count N, the residual stabilizer S_{w-N} grows and the gauge is fixed on a smaller fraction of neurons, so prescribing a fixed N is the wrong scaling; the prescribed fraction, not N, must be held roughly constant. (iii) *Transparency.* The actual-ReLU near-isometry clause relies on the no-clipping event, whose empirical rate is only 49–80% on the deployed architecture (Theorem A.9); deeper or wider stacks lower this rate, so the unconditional L^2 bound (Theorem A.5), not the transparency clause, is what should be leaned on at scale. Raising the no-clipping rate by construction, for example through a small positive bias shift in the middle layers, is an untested extension. These are the failure modes to expect before any foundation-model claim is warranted.

Transformer scale-normalized signature across five sizes. We test the construction on a decoder-only pre-norm transformer with explicit per-head W_Q, W_K, W_V, W_O, on a synthetic Markov / induction-head task (vocab 32+BOS, sequence length 64; 6 seeds, 15 within-scheme pairs per cell), at five sizes spanning $d_{\text{model}} \in \{64, 128, 256\}$, $\#\text{heads} \in \{4, 8, 16\}$, and $L \in \{2, 4\}$ (Table 18). The strong head-prescription (tabulated as HT-strong) prescribes a constant $N = 2$ head roles. Its scale-normalized init distance sits at $d_0/\|W\| \in [0.633, 0.699]$ at every size; the random Kaiming control sits at $[1.349, 1.391]$; a scrambled control (same prescription, head-role assignment randomized per seed to control for magnitude) sits at $[0.768, 0.889]$; an identity-init baseline (tiny weights, residual-preserving) sits at $[0.498, 0.738]$. The $\sim 2\times$ gap between the strong prescription and random in $d_0/\|W\|$ is size-invariant. The identity-Hungarian rate (fraction of pairs whose optimal head permutation is the identity) reports how often the prescription places corresponding head roles in corresponding slots across two seeds. At every size the ratio of strong-prescription over identity-init is 2–3× and the scrambled control matches identity-init, so binding head roles to fixed slots is the operative ingredient, not the magnitude profile. The absolute rate scales with the prescribed fraction $N/\#\text{heads}$: $2/4{=}50\%$ prescribed gives 0.55, $2/8{=}25\%$ gives 0.30, $2/16{=}12.5\%$ gives 0.16. The directional effect is the same at every size; the absolute rate tracks the prescribed fraction. Depth alone does not change the picture: $d_{\text{model}}{=}128$, $L{=}2$ vs. $L{=}4$ and $d_{\text{model}}{=}256$, $L{=}2$ vs. $L{=}4$ give the same signature numbers within seed noise. Whether the absolute identity-Hungarian rate can be held up at larger $\#\text{heads}$ by scaling N in proportion was not tested here remains open.

E.3. Rotational versus permutation symmetry for transformers

Transformers carry two qualitatively different weight-space symmetries, and the construction reaches only one of them. The *discrete* part is head-permutation symmetry: attention heads can be relabeled, exactly the permutation structure our template fixes, and our tiny-transformer pilot canonicalizes it by prescribing fixed head roles (Section H). The *continuous* part is the rotational gauge of the $Q/K/V/O$ subspaces (an orthogonal/general-linear transformation of the per-head key-query and value-output bases that leaves the attention map unchanged), together with LayerNorm scaling freedom. THEORY-INIT as constructed does *not* fix this continuous gauge.

This places continuous-symmetry canonicalization as the natural next target. THEORY-INIT and the W-Asymmetric variant

Table 18. Transformer scale-normalized symmetry-breaking signature across five sizes (decoder-only pre-norm, attention-only). Median over 15 within-scheme pairs per cell (6 seeds). $d_0/\|W\|$ is the init-time Frobenius distance normalized by the mean weight norm of the pair; $\mathcal{B}_{\mathrm{nv}}$ and $\mathcal{B}_{\mathrm{al}}$ are naive and per-layer head-permutation-aligned barriers; id-init and id-train are the fractions of pairs whose Hungarian-optimal head permutation is the identity, at t=0 and after training; cont-frac is the orthogonal-Procrustes $Q/K/V/O$ subspace gauge fraction (Section E.3). HT-strong prescribes $N = 2$ head roles throughout the sweep. The $d_0/\|W\|$ separation between HT-strong ($[0.633, 0.699]$) and random ($[1.349, 1.391]$) is size-invariant, and the identity-Hungarian rate of HT-strong is 2–3× the identity-init baseline at every size. The absolute identity-Hungarian rate of HT-strong tracks $N/\#\text{heads}$: $50\% \to 0.55$, $25\% \to 0.30$, $12.5\% \to 0.16$. A base-size +MLP variant (not shown) gives random 1.349 and HT-strong 1.196: the MLP block adds the magnitude confound that the small-MLP analysis already documents.

size	scheme	d_0	$d_0/\|W\|$	$\mathcal{B}_{\mathrm{nv}}$	$\mathcal{B}_{\mathrm{al}}$	id-init	id-train	cont-frac
d=64, h=4, L=2	random	61.1	1.349	2.244	2.144	0.21	0.22	0.314
	identity	12.0	0.738	0.238	0.232	0.28	0.20	0.153
	HT-strong	16.7	**0.696**	1.153	1.084	**0.58**	**0.55**	0.228
	HT-moder	15.6	0.683	0.941	0.913	0.58	0.54	0.219
	HT-scrambled	21.3	0.889	1.181	1.125	0.28	0.25	0.223
d=128, h=8, L=2	random	106.8	1.368	2.545	2.633	0.14	0.14	0.239
	identity	13.0	0.600	0.182	0.194	0.10	0.12	0.124
	HT-strong	20.2	**0.647**	1.046	1.047	**0.30**	**0.30**	0.190
	HT-moder	18.9	0.628	0.778	0.806	0.30	0.35	0.199
	HT-scrambled	25.7	0.827	0.830	0.966	0.13	0.10	0.193
d=256, h=16, L=2	random	197.6	1.388	2.816	2.743	0.05	0.05	0.175
	identity	16.0	0.535	0.175	0.176	0.05	0.06	0.101
	HT-strong	28.6	**0.672**	0.657	0.683	**0.16**	**0.17**	0.155
	HT-moder	27.1	0.651	0.527	0.521	0.16	0.16	0.151
	HT-scrambled	32.8	0.771	0.592	0.523	0.05	0.07	0.159
d=128, h=8, L=4	random	150.7	1.377	3.129	3.073	0.13	0.14	0.239
	identity	13.9	0.513	0.187	0.186	0.11	0.11	0.125
	HT-strong	22.6	**0.633**	1.111	1.056	**0.23**	**0.23**	0.168
	HT-moder	21.5	0.617	0.807	0.808	0.23	0.24	0.170
	HT-scrambled	27.7	0.775	0.987	0.988	0.15	0.12	0.171
d=256, h=16, L=4	random	278.9	1.391	3.583	3.521	0.06	0.07	0.175
	identity	19.0	0.498	0.184	0.176	0.07	0.06	0.097
	HT-strong	35.2	**0.699**	0.787	0.753	**0.11**	**0.12**	0.136
	HT-moder	34.0	0.686	0.575	0.554	0.11	0.11	0.132
	HT-scrambled	38.7	0.768	0.845	0.527	0.06	0.07	0.136

of Lim et al. (2024) both act on the discrete permutation group, whereas their FiGLU nonlinearity also removes continuous (scaling/diagonal) equivariances; FiGLU is thus a related continuous-symmetry-breaking precedent at the architecture level. The init-time analog we are piloting is a canonical-gauge initialization for the $\mathrm{GL}(r)$ symmetry of a LoRA factorization $\Delta W = BA = (BQ)(Q^{-1}A)$ (Section H): a rank-r SVD of a calibration gradient with a deterministic sign convention fixes the gauge at t=0, with a scrambled-canonical control (random $Q \in \mathrm{GL}(r)$ at fixed ΔW_0) as the negative. Extending the same idea to the attention $Q/K/V/O$ rotational subspaces is open and beyond the scope of the pilot reported here.

Continuous $Q/K/V/O$ subspace gauge probe. To check whether the head-permutation analysis is blind to the continuous gauge, we run an orthogonal Procrustes probe on the head-matched cross-seed pairs of the sweep in Table 18. For each pair (θ_A, θ_B) and each head, we solve R^* for $W_Q \mapsto W_Q R$, $W_K \mapsto W_K R$ (the QK gauge) and S^* for $W_V \mapsto W_V S$, $W_O \mapsto S^\top W_O$ (the OV gauge) in closed form, and report the *continuous-gauge fraction* $\mathrm{cont\_frac} = 1 - \mathrm{residual}/(\text{head-matched raw attention distance})$. A sanity check on a synthetic pure orthogonal-gauge perturbation of one head returns raw distance > 0 and residual $= 0$: the probe finds gauge content exactly when it is there. Across the entire sweep, $\mathrm{cont\_frac}$ sits at 0.10–0.31 and *decreases* with width: at d_{model}=64 (HT-strong, L=2) it is 0.228, then 0.190 at d_{model}=128 (L=2), 0.155 at d_{model}=256 (L=2), 0.168 at d_{model}=128 (L=4), and 0.136 at d_{model}=256 (L=4) (Table 18, last column). The discrete head-permutation analysis is therefore *not* blind to the bulk of the cross-seed disagreement at the scales we tested: 70–90% of the head-matched attention distance is genuine (non-gauge). The continuous symmetries exist, the probe locates them exactly when they are pure, and they simply are not where most of the cross-seed variation lives here. This measures the orthogonal $Q/K/V/O$ subspace gauge only; LayerNorm scale-and-shift gauge and softmax temperature scaling are not probed.

Post-hoc alignment work reaches a complementary conclusion from the trained side. Theus et al. (2025) obtain zero-barrier interpolation between independently trained Vision Transformers and GPT-2 models only after enlarging the alignment search beyond permutations: semi-permutations over heads, invertible maps absorbed into the per-head QK/OV products, and one global orthogonal map on the residual stream. Their analysis locates most of the remaining gap between data-free and task-supervised alignment in that residual-stream orthogonal map, a gauge our probe does not measure. The two observations are consistent: the per-head $Q/K/V/O$ gauge is small at the scales we test, while the residual-stream gauge is the continuous symmetry their results identify as decisive for post-hoc alignment at larger scale. Canonicalizing continuous gauges, starting with the residual stream, is the open question that follows from both lines.

F. Reproducibility

F.1. Hardware and software

Environment. Single workstation with 2× NVIDIA RTX A6000 GPUs, PyTorch with CUDA, RDKit for Crippen atomic contributions on LogP. All experiments use float64 precision throughout. Pinned versions are in the release.

F.2. Protocol, seeds, and released artifacts

For the LMC experiment we train 4 networks per (dataset, init scheme) on a fixed train/validation/test split, with Adam at learning rate 10^{-3}, batch size $\min(256, n_{\text{train}})$, and early stopping on a 20% validation set (patience 200). LMC analysis forms all $\binom{4}{2}$=6 within-scheme and 16 cross-scheme pairs, computes the naive barrier on 21 interpolation points, runs Ainsworth-style iterative weight matching (20 iterations, 5 restarts), and recomputes the aligned barrier and post-alignment Frobenius distance. The 9-method UCI comparison and the LogP novelty ablation use 20 seeds per arm with the analogous protocol. Code, trained checkpoints, and analysis records are available at https://github.com/RandomAnass/theory-init.

Breadth sweep. The depth/width breadth sweep of Table 17 reuses the same training, weight-matching, and interpolation routines, varying only the hidden-width configuration. It covers 4 datasets, 4 hidden configs, 3 methods, 4 seeds, with the matching protocol above.

G. Limitations

G.1. Architecture and scale

THEORY-INIT as constructed applies to feedforward ReLU MLPs at modest width and depth ($w = 64$, $L = 4$, $\sim$ 9k parameters; the molecular LogP accuracy/novelty runs use $w = 128$, while the LMC LogP networks use $w = 64$ like the other datasets). The Stage 2 PWL physics-neuron construction is specific to MLPs. Extension to convolutional architectures works on MNIST and, conditionally on per-channel filter coverage, on CIFAR-10 (Section D.7), with the size of the effect tracking the prescribed first layer's coverage of the relevant input axes (RGB on CIFAR-10). The scalability pilot tested five transformer sizes ($d_{\text{model}} \in \{64, 128, 256\}$, $\#\text{heads} \in \{4, 8, 16\}$, $L \in \{2, 4\}$): the scale-normalized cross-seed distance signature held at every size, and the absolute identity-Hungarian rate tracked the prescribed/total head ratio rather than absolute scale (Table 18). On the CNN side, the depth + residual pilot (Table 16) transmits the per-channel prescription through $D \in \{2, 3, 4, 5\}$ convs above conv1 and through a residual variant: $d_{\text{norm}}(\text{THEORY-INIT})$ stays below the random-vector baseline at every depth and below the magnitude-matched scrambled control at every depth. Extension to attention architectures at real-language scale remains untested.

G.2. Train-MSE / distance confound

Within THEORY-INIT, runs that achieve lower train MSE also tend to be closer in weight space ($\rho \approx 0.95$ within-scheme; $\rho \approx 0.81$ within He). Better-fit networks occupy a tighter region of weight space, so part of THEORY-INIT's pre-alignment effect could be inherited from THEORY-INIT's accuracy advantage rather than from the prescribed Stage-2 row assignment. A direct disentanglement test: we replace THEORY-INIT's prescribed Stage-1 rows with random Gaussian unit vectors of matched row-norm profile (deterministic across the pair, per-seed He padding; max row cosine similarity vs. correct theory ≤ 0.04 on LogP). Across all four datasets this control reproduces THEORY-INIT's symmetry-breaking signature: d_0 matches to within sampling noise (Yacht 1.96 vs 1.96; Energy 2.00 vs 2.04; Concrete 2.01 vs 2.07; LogP 4.80 vs 4.86), and identity-Hungarian rate and barrier reduction are comparable. Test MSE, however, depends on the dataset: on Yacht random-Gaussian achieves 0.0117 vs THEORY-INIT's 0.0010 ($11\times$ worse) but is still $2.5\times$ better than He (0.0293); on Energy, Concrete, and Lipophilicity random-Gaussian is within 1–4% of THEORY-INIT. Combined with the LogP novelty ablation (Crippen as residual / feature / frozen all within $\pm 1.3\%$ of He, while THEORY-INIT achieves -7.0%, Section D.2), the symmetry-breaking signature is scaffold-driven (deterministic + magnitude-matched Stage-1 rows + Stage-2 $I + \Delta$ + Stage-3 small-random), while theory correctness contributes an accuracy advantage that depends on how well the prescribed rows align with the dimensions the trained network actually uses. We do not claim this generalizes beyond our four datasets, and the appendix's other controls (Equation (28), the N-scan, the tiny-He magnitude baseline) similarly leave a complete causal decomposition of accuracy, basin geometry, and initialization-induced alignment open.

G.3. LMC barrier vs trained-loss ratio

None of the schemes here exhibit linear mode connectivity in the strict sense. Naive barriers exceed the trained test MSE by 1–2 orders of magnitude (Table 10), dominated by the high-loss intermediate alphas where small MLPs sit far from any minimum. What we report, conditional on this barrier-violation regime, is that the violation magnitude is consistently smaller than the He baseline (Table 10). The asymmetry between init schemes is interesting; LMC *per se* is not.

G.4. LogP layer-wise matching instability

On LogP and Concrete, layer-wise weight matching is unstable in the He–theory cross-scheme regime: the aligned barrier exceeds the naive barrier in some pairs. This matches prior reports that alignment methods alone are insufficient for low-barrier interpolation in low-overparameterization settings (Ainsworth et al., 2023; Jordan et al., 2023). For THEORY-INIT–THEORY-INIT pairs the issue does not arise (aligned barrier $\leq$ naive), consistent with Theorem A.21 predicting identity is already optimal. Activation-matching variants might give cleaner cross-scheme alignment but were not tested.

G.5. Where the theory applies

The signal-preservation theorem (Prop. A.2) and the trace-dominance lemma (Theorem A.19) are stated under the small-MLP default architecture and theory regime (small layer-1 padding, $\sigma \leq 10^{-2}$ middle-layer noise). The PSD/trace argument requires only that the deterministic theory block $G = TT^\top + \beta\beta^\top$ be PSD, which is automatic; sharpness depends on the spectral gap of G and is verified numerically on each dataset (Theorem A.20). Strict-positive variants of the architecture (e.g., padding with a small bias before ReLU, or softplus padding rows) would raise the empirical transparency rate of Theorem A.9 to near 100% and make the actual-ReLU clause of Theorem A.12 nearly-sure; this comparison is open.

H. Future directions

The symmetry-breaking template above is specific to permutation symmetries of feedforward ReLU MLPs. Two extensions to other weight-space symmetry groups are in progress; we report preliminary findings here only to indicate scope.

Head-permutation symmetry in tiny transformers. We are extending the template to attention symmetries. In a 2-layer decoder-only transformer (d_{model}=64, 4 heads $\times d_{\mathrm{head}}$=16, $\sim$30k attention-only parameters) trained on synthetic Markov / induction-head sequences, prescribing fixed head roles (previous-token, copy, induction) at initialization yields the same canonical-orbit-selection signature on the head-permutation subgroup. Across 4 seeds in the attention-only regime the prescribed init is $\sim$3.7$\times$ closer at t=0 in $\|\theta\|_F$ than random, naive barriers are 40–50$\%$ smaller, per-layer head-permutation Hungarian solving returns the identity at twice the random rate, and a scrambled-head control (same construction, role assigned to a random head per seed) destroys the effect. This pilot has since been swept across five sizes ($d_{\mathrm{model}} \in \{64, 128, 256\}$, $\#\mathrm{heads} \in \{4, 8, 16\}$, $L \in \{2, 4\}$) and complemented by an orthogonal $Q/K/V/O$ subspace gauge probe; the scale-normalized signature holds at every size, the identity-Hungarian rate tracks the prescribed/total head ratio, and the continuous gauge fraction is small and decreases with width (Sections E.2 and E.3 and Table 18). A complete treatment, including real language tasks and the additional LayerNorm and softmax-scaling gauges, is beyond the scope of this workshop paper.

Gauge-invariant LoRA initialization. Low-rank adapters $\Delta W = BA$ admit a continuous $\mathrm{GL}(r)$ gauge symmetry $BA = (BQ)(Q^{-1}A)$ that random and gradient-SVD-style initializations leave unfixed. We are investigating canonical-gauge initialization (rank-r SVD of a calibration gradient with deterministic sign convention), with a scrambled-canonical control that applies a random $Q \in \mathrm{GL}(r)$ while preserving $\Delta W_0 = BA$. Preliminary experiments fine-tuning a small instruction-following adapter (r=8, 4 seeds) show $32\times$ smaller cross-seed factor-space distance and $33\times$ smaller gauge-invariant product-space distance compared to vanilla LoRA, while the scrambled-canonical control yields 5–6$\times$ higher product distance despite an identical ΔW_0.

Close-but-not-exact theory. When the theory-derived prior is itself an approximation of the target, THEORY-INIT still encodes useful inductive bias. We illustrate this on two physically motivated correction problems (Figure 8): Black–Scholes $\rightarrow$ Heston (constant- vs. stochastic-volatility option prices) and Euler–Bernoulli $\rightarrow$ Timoshenko (slender- vs. thick-beam deflections). Initializing with the closed-form approximant gives near-identical start MSE to a He-trained warm start, yet drops 3 orders of magnitude further at convergence (final-MSE separations $\sim 6\times$ on Heston, $\sim 10^2$–$10^3\times$ on Timoshenko under identical Adam protocols, 4 seeds, $2{,}000$ epochs). The 3D landscape view shows that the two methods, despite similar initial loss, sit in qualitatively different basins separated by a ridge: the THEORY-INIT basin is what survives loss-of-plasticity in this regime, not just the point-prior.

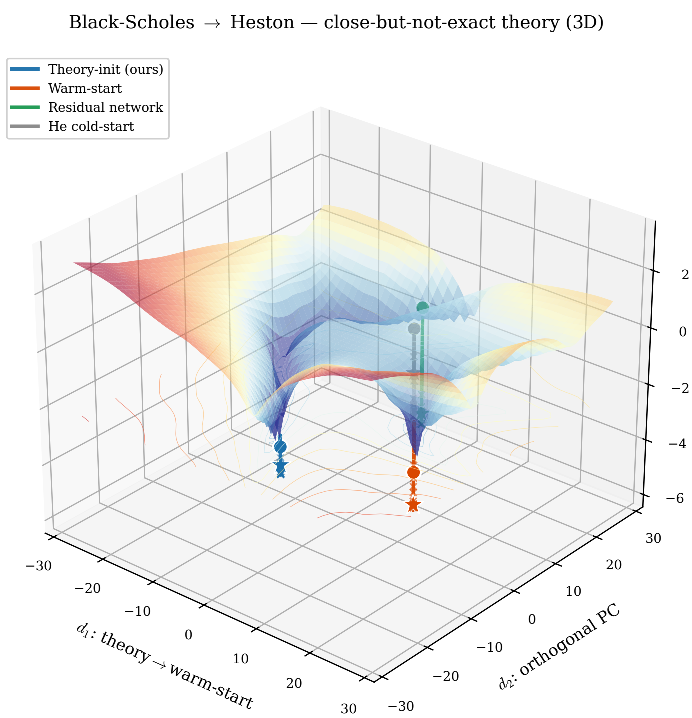

Figure 8. Close-but-not-exact THEORY-INIT: 3D loss landscape for Black–Scholes → Heston, 51×51 grid in the (theory→warm, top PCA) plane. THEORY-INIT descends into a deeper, narrower basin than warm-start; trajectories shown for THEORY-INIT, warm-start, residual-network, He-cold. Companion 2D contour and Euler–Bernoulli → Timoshenko sanity scenario are in the public release.